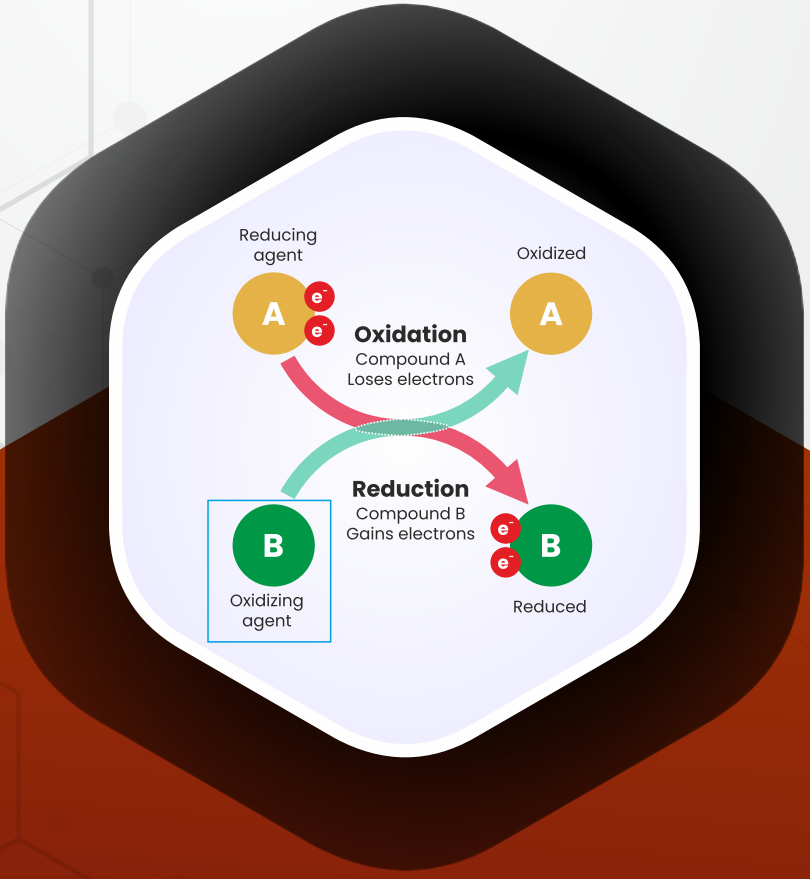
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Redox & Equivalent Concepts

(English Medium)

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REDOX REACTIONS & EQUIVALENT CONCEPTS

1. OXIDATION & REDUCTION

Let us do a comparative study of oxidation and Reduction ;

	Oxidation		**Reduction**
(1)	Addition of oxygen $Mg \rightarrow MgO$	(1)	Removal of oxygen $CuO \rightarrow Cu$
(2)	Removal of Hydrogen $H_2S \rightarrow S$	(2)	Addition of Hydrogen $S \rightarrow H_2S$
(3)	Addition of electronegative atom or group $Mg \rightarrow MgCl_2$	(3)	Removal of Electronegative atom or group $FeCl_3 \rightarrow FeCl_2$
(4)	Removal of electropositive atom or group $K_4[Fe(CN)_6] \rightarrow K_3[Fe(CN)_6]$	(4)	Addition of electropositive atom or group $Cl_2 \rightarrow MgCl_2$
(5)	Increase in positive charge $Fe^{2+} \rightarrow Fe^{3+}$	(5)	Decrease in positive charge $Fe^{3+} \rightarrow Fe^{2+}$
(6)	Increase in oxidation number +2 +4 $SnCl_2 \rightarrow SnCl_4$	(6)	Decrease in oxidation number +7 $MnO_4^- \rightarrow Mn^{2+}$
(7)	Loss of electron $Sn^{2+} \rightarrow Sn^{4+} + 2e^-$	(7)	Gain of electron $Fe^{3+} + e^- \rightarrow Fe^{2+}$

2. OXIDATION NUMBER OR OXIDATION STATE :

◆ Oxidation number denotes the oxidation state of an element in a compound ascertained according to a set of rules formulated on the basis that electron pair in a covalent bond belongs entirely to more electronegative element.

For example , in CO_2 , the oxidiation state of carbon is +4 and it is also its oxidation number. Simiarlary, the oxidation state as well as the oxidation number of oxygen is –2. This implies that the oxidation number denotes the oxidation state of an element in a compound.

Now, understand the determination of oxidation number with the help of these examples :

(a) **HCl**

Structural formula : H – Cl

As chlorine is more electronegative than hydrogen, counting shared electron pair for chlorine will result $H^+ Cl^-$ and hence,

oxidation number of H = +1

oxidation number of Cl = –1

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(b) **CO_2**

Structural formula : O = C = O

As oxygen is more electronegative than carbon, counting shared electron pair for oxygen with result $O^{2-}\ {}^{2+}C^{2+}\ {}^{2-}O$ and hence,

Oxidation number of C = +4

Oxidation number of O = –2

2.1 Rules to determine oxidation number :

The following rules are helpful in calculating oxidation number of the elements in their different compounds. It must be remembered that the basis of these rule is the electronegativity of the element.

(i) Oxidation number of an element in free state or in allotropic forms is always zero.

e.g. $\overset{0}{O_2}, \overset{0}{S_8}, \overset{0}{P_4}, \overset{0}{O_3}, \overset{0}{Na}$

(ii) Sum of the oxidation number of all the atoms in a molecule is zero.

(iii) Sum of the oxidation number of all the atoms in an ions is equal to the charge on the ion.

(iv) If the group number of an element in periodic table is 'n' (I, II, III, etc) then its oxidation number may vary from (+n) to (n – 8).

e.g. N-atom belongs to V group in the periodic table therefore as per rule its oxidation number may vary from –3 to +5 $(\overset{-3}{NH_3}, \overset{+2}{NO}, \overset{+3}{N_2O_3}, \overset{+4}{NO_2}, \overset{+5}{N_2O_5})$

Exceptions :

Cu(IB) : + 1, +2 ;

Au (IB) : +1, +3 , etc

(v) **Metals :**

(a) Alkali metal (Li, Na, K, Rb, Cs, Fr) always have oxidation number +1 in all of their compounds.

(b) Alkaline earth metal (Be, Mg, Ca , Sr, Ba, Ra) always have oxidation number +2 in all of their compounds.

(c) Aluminium always have +3 oxidation number in all of its compounds.

(vi) **Hydrogen atom :**

H atom has oxidation number equal to +1 in all of its compound, but in metallic hydrides (e.g. NaH, KH), it is –1.

(vii) **Oxygen atom :**

Oxygen atom has oxidation number equal to –2 in all of its compounds except

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(a) peroxides (e.g. H_2O_2, Na_2O_2) , where it is –1

(b) super oxides (e.g. KO_2) where it is $-\frac{1}{2}$

(c) ozonides (KO_3), where it is $-\frac{1}{3}$

(d) oxygen fluorides (O_nF_2), where it is $\left(+\frac{2}{n}\right)$.

(viii) Halogen atom :

Fluorine is most electronegative atom. It always has oxidation no. equal to –1 in all of its compound.

All other halogen atoms (Cl, Br, I) has oxidation number equal to –1 in all of their compounds. But if halogen atom is attached with an more electronegative atom, than halogen atom then it will show positive oxidation numbers.

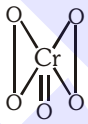
Eg. $\overset{+5}{K}ClO_3$, $\overset{+5}{H}IO_3$, $\overset{+7}{H}ClO_4$, $\overset{+5}{K}BrO_3$, etc

(ix) If two or more than two atoms of an element are present in the molecule or ion such as $Na_2S_2O_3$, $Cr_2O_7^{2-}$, etc, the oxidation number of the atom of that element will then be the average of the oxidation number of all the atoms of that element.

2.2 SOME SPECIAL CASES :

In order to determine the exact or individual oxidation number we need to take help from the structures of the molecules. Some special cases are discussed as follows :

(i) CrO_5

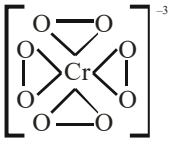
The structure of CrO_5 is 

From the structure it is evident that in CrO_5 there are two peroxide linkages The contribution of each peroxide linkage is –2. Let the O.N. of Cr is x.

$\therefore$ $x + (-2)2 + (-2) = 0$

$\therefore$ O.N. of Cr = x = +6

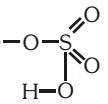
(ii) K_3CrO_8

The structure of CrO_8^{3-} is 

From the structure it is evident that in CrO_8^{-3} there are four peroxide linkages. The contribution of each peroxide linkage is –2. Let the O.N. of Cr is x.

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(iii) H_2SO_5

The structure of H_2SO_5 is 

From the structure, it is evident that in H_2SO_5, there are one peroxide linkage , Let the O.N. of S = x

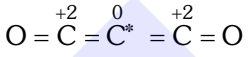
$\therefore$ $+1 - 2 + x + (-2)\,2 + (-2) + 1 = 0$

$\therefore$ O.N. of S = x = 6

2.3 PARADOX OF FRACTIONAL OXIDATION NUMBER

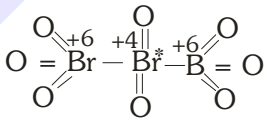
Fractional oxidation state is the average oxidation state of the element under examination and the structural parameters reveal that the element for whom fractional oxidation state is realised is actually present in different oxidation states. Structure of the species C_3O_2, Br_3O_8 and $S_4O_6^{2-}$ reveal the following bonding situations. The element marked with asterisk in each species is exhibiting the different oxidation state (oxidation number) from rest of the atoms of the same element in each of the species.

- This reveals that in C_3O_2, two carbon atoms are present in +2 oxidation state each whereas the third one is present in zero oxidation state and the average is 4/3. However the realistic picture is +2 for two terminal carbons and zero for the middle carbon.



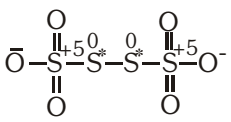
Structure of C_3O_2
(Carbon suboxide)

- Likewise in Br_3O_8, each of the two terminal bromine atoms are present in +6 oxidation state and the middle bromine is present in +4 oxidation state. Once again the average, that is different from reality is 16/3.



Structure of Br_3O_8 (tribromooctaoxide)

- In the same fashion, in the species $S_4O_6^{2-}$, is 2.5, whereas the reality being +5,0,0 and +5 oxidation number respectively for each sulphur.



Structure of $S_4O_6^{2-}$ (tetrathionate ion)

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Ex.1 Calculate oxidation number of underlined element in the following:

(i) $Na_2\underline{S}_2O_3$ (ii) $Na_2\underline{S}_4O_6$ (iii) $\underline{Cl}O^-$, (iv) $\underline{N}O_2^-$,

(v) $\underline{N}O_3^-$ (vi) $\underline{C}Cl_4$ (vii) $K_2\underline{Cr}O_4$ (viii) $K\underline{Mn}O_4$

Sol. (i) Let oxidation number of S-atom is x.

$(+1) \times 2 + (x) \times 2 + (-2) \times 3 = 0$

$x = +2$

(ii) Let oxidation number of S-atom is x

$(+1) \times 2 + (x) \times 2 + (-2) \times 6 = 0$

$x = +2.5$

(iii) $x - 2 = -1$ (where 'x' is O.N. of Cl)

$\therefore x = +1$

(iv) $2 \times (-2) + x = -1$ (where 'x' is O.N. of N)

$\therefore x = +3$

(v) $x + 3 \times (-2) = -1$ (where 'x' is O.N. of N)

$x = +5$

(vi) $x + 4 \times (-1) = 0$ (where 'x' is O.N. of C)

$\therefore x = +4$

(vii) $2 \times +1 + x + 4 \times (-2) = 0$ (where 'x' is O.N. of Cr)

$\therefore x = +6$

(vi) $+1 + x + 4 \times (-2) = 0$ (where x is O.N. of Mn).

$\therefore x = +7$

Ex.2. Identify the species undergoing oxidation and reduction in the following reactions :

(i) $N_2 + H_2 \rightarrow NH_3$

(ii) $Zn + HCl \rightarrow ZnCl_2 + H_2$

(iii) $Mg + O_2 \rightarrow MgO$

(iv) $Zn + NaOH \rightarrow Na_2ZnO_2 + H_2$

(v) $H_2SO_4 + Mg \rightarrow MgSO_4 + SO_2 + H_2O$

(vi) $H_2SO_4 + NaOH \rightarrow Na_2SO_4 + H_2O$

(vii) $Cl_2 + NaOH \rightarrow NaCl + NaClO_3 + H_2O$

(viii) $FeCl_3 + SnCl_2 \rightarrow FeCl_2 + SnCl_4$

(ix) $AgNO_3 + NaCl \rightarrow AgCl_2 + NaNO_3$

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Sol. *(i)* $\underset{(R)}{\overset{(O)}{\overbrace{N_2 + \underbrace{H_2 \longrightarrow NH_3}}}}$

(ii) $\underset{(O)}{\overset{(R)}{\overbrace{Zn + \underbrace{HCl \longrightarrow ZnCl_2 + H_2}}}}$

(iii) $\overset{(R)}{\overbrace{\underbrace{Mg + O_2 \longrightarrow MgO}}}$

(iv) $\underset{(R)}{\overset{(O)}{\overbrace{Zn + \underbrace{NaOH \longrightarrow Na_2ZnO_2 + H_2}}}}$

(v) $\underset{(O)}{\overset{(R)}{\overbrace{H_2SO_4 + \underbrace{Mg \longrightarrow MgSO_4 + SO_2 + H_2O}}}}$

(vi) $H_2SO_4 + NaOH \longrightarrow Na_2SO_4 + H_2O$
Neither O nor R

(vii) $\underset{(O)}{\overset{(R)}{\overbrace{Cl_2 + \underbrace{NaOH \longrightarrow NaCl + NaClO_3} + H_2O}}}$

(viii) $\underset{(O)}{\overset{(R)}{\overbrace{FeCl_3 + \underbrace{SnCl_2 \longrightarrow FeCl_2 + SnCl_4}}}}$

(ix) $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$
Neither O *nor* R

Ex.3. Identify the oxidising and reducing agents in the following reactions :

(i) $Fe(s) + \frac{1}{2}O_2(g) \rightarrow FeO(s)$

(ii) $2FeO(s) + \frac{1}{2}O_2(g) \rightarrow Fe_2O_3(s)$

(iii) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

(iv) $2MnO_4^-(aq.) + 16H^+(aq.) + 10\,I^-(aq.) \rightarrow 2Mn^{2+}(aq) + 5I_2(s) + 8H_2O(l)$

(v) $3Cu(s) + 8H^+(aq.) + 2NO_3^-(aq.) \rightarrow 3Cu^{2+}(aq.) + 2NO(g) + 4H_2O(l)$

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Sol. (i) $\overset{0}{Fe}(s) + \frac{1}{2}\overset{0}{O_2}(g) \rightarrow \overset{+2}{Fe}\overset{-2}{O}(s)$

R.A. O.A.

(ii) $2\overset{+2}{Fe}O(s) + \frac{1}{2}\overset{0}{O_2}(g) \rightarrow \overset{+3}{Fe_2}\overset{-2}{O_3}(g)$

R.A. O.A.

(iii) $\overset{0}{N_2}(g) + 3\overset{0}{H_2}(g) \rightarrow 2\overset{-3+1}{NH_3}(g)$

O.A. R.A.

(iv) $2\overset{+7}{MnO_4^-}(aq.) + 16\,H^+(aq.) + 10\,I^-(aq.) \rightarrow 2Mn^{2+}(aq.) + 5\overset{0}{I_2}(s) + 8H_2O(l)$

O.A. R.A.

(v) $2\overset{0}{Cu}(s) + 8\,H^+(aq.) + 2\overset{+5}{NO_3^-}(aq.) \rightarrow 3Cu^{2+}(aq.) + 2\,NO^{+2}(g) + 4H_2O(l)$

R.A. O.A.

3 OXIDISING AND REDUCING AGENTS :

3.1 Oxidising agent or Oxidant :

Oxidising agents are those compound which can oxidise others and reduced itself during the chemical reaction. Those reagents whose O.N. decrease or which gain electron(s) in a redox reaction are termed as oxidants.

e.g. $KMnO_4$, $K_2Cr_2O_7$, HNO_3, conc. H_2SO_4 etc, are some powerful oxidising agents.

3.2 Reducing agent or Reductant :

Reducing agents are those compound which can reduce others and oxidise itself during the chemical reaction. Those reagents whose O.N. increase or which loses electron in a redox reaction are termed as reductants.

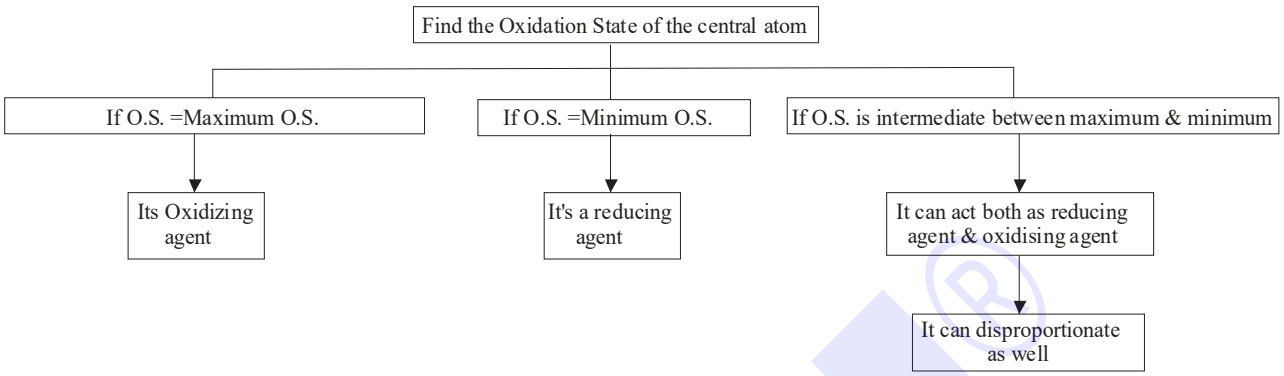
e.g. KI, $Na_2S_2O_3$, etc are some powerful reducing agents.

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Note : There are some compounds also which can work both oxidising agent and reducing agent.

e.g. H_2O_2, NO_2^-

3.3 How to identify whether a particular substance is an oxidising or reducing agent :



4 REDOX REACTIONS

A reaction in which oxidation and reduction simultaneously take place.

In all redox reactions, the total increase in oxidation number must equal the total decrease in oxidation number.

e.g. $10\overset{+2}{Fe}SO_4 + 2K\overset{+7}{Mn}O_4 + 8H_2SO_4 \longrightarrow 5\overset{+3}{Fe_2}(SO_4)_3 + 2\overset{+2}{Mn}SO_4 + K_2SO_4 + 8H_2O$

TYPE OF REDOX REACTION

4.1 Combination Reactions :

When two or more element combine to form compound, then such combination reactions are redox reaction.

For example : $A + B \longrightarrow C$

(a) $3\overset{0}{Mg}(s) + \overset{0}{N_2}(g) \xrightarrow{\Delta} \overset{+2}{Mg_3}\overset{-3}{N_2}(s)$

(b) $\overset{0}{C}(s) + \overset{0}{O_2}(g) \xrightarrow{\Delta} \overset{+4}{C}\overset{-2}{O_2}(g)$

(c) $3\overset{0}{Li}(s) + \overset{0}{N_2}(g) \xrightarrow{\Delta} \overset{+1}{Li_3}\overset{-3}{N}(s)$

4.2 Decomposition Reactions :

Decomposition reactions are opposite of combination. Decomposition reactions are those in which when a compound breakdown to form two or more components. If at least one of the product is in the elemental state, such decomposition reactions are definitely redox. For example :

(a) $2\overset{+1}{K}\overset{+5}{Cl}\overset{-2}{O_3}(s) \xrightarrow{\Delta} 2\overset{+1}{K}\overset{-1}{Cl}(s) + 3\overset{0}{O_2}(g)$

(b) $\overset{+2}{Ca}\overset{-1}{H_2} \xrightarrow{\Delta} \overset{0}{Ca}(s) + \overset{0}{H_2}(g)$

(c) $2\overset{+1}{Na}\overset{-1}{H}(s) \xrightarrow{\Delta} 2\overset{0}{Na}(s) + \overset{0}{H_2}(g)$

(d) $\overset{+1}{H_2}\overset{-2}{O}(\ell) \xrightarrow{\Delta} 2\overset{0}{H_2}(g) + \overset{0}{O_2}(g)$

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If must be noted that all decomposition reaction are not redox:

$$\overset{+2}{Ca}\overset{+4}{C}\overset{-2}{O_3}(s) \xrightarrow{\Delta} \overset{+2}{Ca}\overset{-2}{O}(s) + \overset{+4}{C}\overset{-2}{O_2}(g)$$

4.3 Displacement Reactions :

A reaction in which an atom or ion in a compund is replaced by an atom or ion of another element is called a displacement reaction. It may be denoted as :

$$X + YZ \longrightarrow XZ + Y$$

Displacement reactions fit into two categories :

4.3.1 Metal displacement reactions : A metal in a compound can be displaced by another metal in the uncombined state.

$$\overset{+2\ +6\ -2}{CuSO_4}(aq) + \overset{0}{Zn}(s) \rightarrow \overset{0}{Cu}(s) + \overset{+2\ +6\ -2}{ZnSO_4}(aq)$$

$$\overset{+5\ -2}{V_2O_5}(s) + \overset{0}{5Ca}(s) \xrightarrow{\Delta} \overset{0}{2V}(s) + \overset{+2\ -2}{5CaO}(s)$$

$$\overset{+4\ -1}{TiCl_4}(l) + \overset{0}{2Mg}(s) \xrightarrow{\Delta} \overset{0}{Ti}(s) + 2\overset{+2\ -1}{MgCl_2}(s)$$

$$\overset{+3\ -2}{Cr_2O_3}(s) + \overset{0}{2\ Al}(s) \xrightarrow{\Delta} \overset{+3\ -2}{Al_2O_3}(s) + \overset{0}{2Cr}(s)$$

4.3.2 Non-metal displacement reactions : The non-metal displacement redox reactions include hydrogen displacement (most common), halogen displacement and rarely occurring reaction involving oxygen displacecment.

$$\overset{0}{2Na}(s) + \overset{+1\ -2}{2H_2O}(l) \rightarrow \overset{+1\ -2\ +1}{2NaOH}(aq) + \overset{0}{H_2}(g)$$

$$\overset{0}{Ca}(s) + \overset{+1\ -2}{2H_2O}(l) \rightarrow \overset{+2\ -2\ +1}{Ca(OH)_2}(aq) + \overset{0}{H_2}(g)$$

Less active metals such as magnesium and iron react with steam to produce dihydrogen gas:

$$\overset{0}{Mg}(s) + \overset{+1\ -2}{2H_2O}(l) \xrightarrow{\Delta} \overset{+2\ -2\ +1}{Mg(OH)_2}(s) + \overset{0}{H_2}(g)$$

$$\overset{0}{2Fe}(s) + \overset{+1\ -2}{3H_2O}(l) \xrightarrow{\Delta} \overset{+3\ -2}{Fe_2O_3}(s) + \overset{0}{3H_2}(g)$$

Many metals, including those which do not react with cold water, are capable of displacing hydrogen from acids. Dihydrogen from acids may even be produced by such metals which do not react with steam.

$$\overset{0}{Zn}(s) + \overset{+1\ -1}{2HCl}(aq) \rightarrow \overset{+2\ -1}{ZnCl_2}(aq) + \overset{0}{H_2}(g)$$

$$\overset{0}{Mg}(s) + \overset{+1\ -1}{2HCl}(aq) \rightarrow \overset{+2\ -1}{MgCl_2}(aq) + \overset{0}{H_2}(g)$$

Fluorine is so reactive that it can replace chloride, bromide and iodide ions in solution. In fact, fluorine is so reactive that it attacks water and displaces the oxygen of water :

$$\overset{+1\ -2}{2H_2O}(l) + \overset{0}{2F_2}(g) \rightarrow \overset{+1\ -1}{4HF}(aq) + \overset{0}{O_2}(g)$$

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It is for this reason that the displacement reactions of chlorine, bromine and iodine using fluorine are not generally carried out in aqueous solution. On the other hand, chlorine can displace bromide and iodide ions in an aqueous solution as shown below:

$$\overset{0}{Cl_2}(g) + 2\overset{+1\ -1}{KBr}(aq) \rightarrow 2\,KCl(aq) + \overset{0}{Br_2}(l)$$

$$\overset{0}{Cl_2}(g) + 2\overset{+1\ -1}{KI}(aq) \rightarrow 2\,KCl(aq) + \overset{0}{I_2}(s)$$

4.4 Disproportionations reactions :

A redox reaction in which the same element present in a particular compound (molecule or ion) in a definite oxidation state is oxidized as well as reduced simultaneously is a disproportionation reactions.

Disproportionation reactions are a special type of redox reactions. One of the reactants in a disproportionation reaction always contains an element that can exist in at least three oxidation states. The element in the form of reacting substance is in the intermediate oxidation state and both higher and lower oxidation states of that element are formed in the reaction.

$$2\overset{+1\ -1}{H_2O_2}(aq) \longrightarrow 2\overset{-2}{H_2O}(\ell) + \overset{0}{O_2}(g)$$

$$\overset{0}{S_8}(s) + 12OH^-(aq) \longrightarrow 4\overset{-2}{S^{2-}}(aq) + 2\overset{+2}{S_2O_3^{2-}}(aq) + 6H_2O(\ell)$$

$$\overset{0}{Cl_2}(g) + 2OH^-(dil.) \longrightarrow \overset{+1}{ClO^-}(aq) + \overset{-1}{Cl^-}(aq) + H_2O(\ell)$$

$$3\overset{0}{Cl_2}(g) + 6OH^-(conc.) \longrightarrow \overset{+5}{ClO_3^-}(aq) + 5\overset{-1}{Cl^-}(aq.) + 3H_2O(\ell)$$

- **Consider the following reactions :**

(a) $2KClO_3 \rightarrow 2KCl + 3O_2$

$KClO_3$ plays a role of oxidant and reductant both. Because same element is not oxidised and reduced. Here, Cl present in $KClO_3$ is reduced and O present in $KClO_3$ is oxidized. So **it is not a disproportion reaction** although it looks like one.

(b) $NH_4NO_2 \longrightarrow N_2 + 2H_2O$

Nitrogen in this compound has –3 and +3 oxidation number but belongs to different ions so it is not a disproportion reaction.

(c) $4KClO_3 \longrightarrow 3KClO_4 + KCl$

It is case of disproportionation reaction in which Cl is disproportionating.

Note : **F_2 does not undergo disproportionation as it is the most electronegative element.**

$$2F_2 + 2NaOH\ (dil) \longrightarrow 2NaF + OF_2 + H_2O$$

$$2F_2 + 4NaOH\ (conc.) \longrightarrow 4NaF + O_2 + 2H_2O$$

4.5 Comproportionation reaction :

Reverse of disproportionation is called comproportionation. It is a class of redox reaction in which an element from two different oxidation state gets converted into a single oxidation state. In some of the disproportionation reactions by changing the medium (from acidic to basic or reverse) the reaction goes in backward direction and can be taken as an example of comproportionation.

$$I^- + IO_3^- + H^+ \longrightarrow I_2 + H_2O\ (acidic)$$

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4.6 Intermolecular and intramolecular redox reactions :

In **intermolecular redox reaction**, oxidation and reduction of different species occur. For example :

$$\overset{-2}{2H_2S} + \overset{+4}{SO_2} \longrightarrow 3\overset{0}{S} + 2H_2O$$

$$\overset{0}{Zn} + 2\overset{+1}{HCl} \longrightarrow \overset{+2}{ZnCl} + \overset{0}{H_2}$$

In **Intramolecular redox reaction ,** the same species get oxidised as well as reduced. For example:

$$\overset{+5\ -2}{2KClO_3} \longrightarrow \overset{-1}{2KCl} + 2\overset{0}{O_2}$$

Oxidation and reduction occurred for the same compound , $KClO_3$.

Disproportionation reaction is a type of intramolecular redox reaction, in which oxidation and reduction occur due to the same element, from same molecule or ion.

5. NON-REDOX REACTIONS :

These are the reactions in which neither oxidation non reduction take place. Some example of non-redox reactions are :

(i) Combination reactions like :

$NH_3 + HCl \rightarrow NH_4Cl$

$SO_3 + H_2O \rightarrow H_2SO_4$

(ii) Decomposition reactions like :

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$$

(iii) Neutralisation reactions like :

$HCl + NaOH \rightarrow NaCl + H_2O$

$H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$

(iv) Precipitation reactions like :

$AgNO_3(aq.) + NaCl(aq.) \rightarrow AgCl(s)\downarrow + NaNO_3(aq.)$

$BaCl_2(aq.) + Na_2SO_4(aq.) \rightarrow BaSO_4(s)\downarrow + 2NaCl(aq.)$

6. BALANCING OF REDOX REACTION

All balanced equations must satisfy two criteria :

(i) Atom balance (mass balance) :

There must be the same number of atoms of each element in reactant and product side.

(ii) Charge balance :

The sum of actual total charges on both side of the equation must be equal.

There are two methods for balancing the redox reactions

(a) Oxidation - number method

(b) Ion electron method or half cell method

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6.1 Oxidation number method :

This method was given by Jonson. In a balanced redox reaction, **total increase in oxidation number must be equal to total decreases in oxidation number.** This equivalence provides the basis for balancing redox reactions. The general procedure involves the following steps :

Step-I : Write the correct formula for each reactant and product.

Step-II : Identify atoms which undergoes change in oxidation number in the reaction by assigning the oxidation number to all elements in the reaction.

Step-III : Balance the number of atoms of the elements by simple counting, for which changes in oxidation numbers are taking place.

Step-IV : Calculate the increase or decrease in the oxidation number per atom and for the entire molecule/ion in which it occurs. If these are not equal, then multiply by suitable number so that they become equal.

Step-V : Balance the atoms of all other elements (for which change in oxidation number was not occuring), by simple counting (adjusting the stoichiometric coeffcients of other compounds with the help of stoichiometric coefficients of the components fixed in step-III).

Step-VI : If the given reaction is in ionic form, ascertain that the total charges in reactants and products are equal. Addition of H^+ or OH^- ions and / or H_2O molecule may be helpful in balancing charge and even, atoms in step- IV. For reaction in acidic medium, H^+ ions and for reaction in basic medium, OH^- ions are used, as required.

Ex.4 Balance the following chemical reaction by oxidation number method and write their skeleton equation :

The nitrate ions in acidic medium oxidize magnesium to Mg^{2+} ions but itself gets reduced to nitrous oxide.

Sol. The skeleton equation for the given process is

$$Mg + NO_3^- \longrightarrow Mg^{2+} + N_2O + H_2O$$

Step (I) : $\overset{0}{Mg} + (\overset{+5\ -2}{NO_3})^- \longrightarrow Mg^{2+} + \overset{+1\ -2}{N_2O} + \overset{+1\ -2}{H_2O}$

Multiply NO_3^- by 2 to equalize N atoms

Step (II) : O.N. increases by 2 per Mg atom

$$\overset{0}{Mg} + 2NO_3^- \longrightarrow Mg^{2+} + \overset{+1}{N_2}O + H_2O$$

$4 \times 2 = 8$

O.N. decreases by 4 per N atom

Step (III) : Equalize increase/decrease in O.N. by multiplying Mg by 4 and $2NO_3^-$ by 1.

$$4Mg + 2NO_3^- \longrightarrow 4Mg^{2+} + N_2O + H_2O$$

Step (IV) : Balance O atoms

$$4Mg + 2NO_3^- \longrightarrow 4Mg^{2+} + N_2O + H_2O + 4H_2O$$

Step (V) : Balance H atoms as is done in acidic medium.

$$4Mg + 2NO_3^- + 10H^+ \longrightarrow 4Mg^{2+} + N_2O + 5H_2O$$

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Ex5. Balance the following redox reaction oxidation number method.

$$KMnO_4 + H_2SO_4 + H_2C_2O_4 \longrightarrow K_2SO_4 + MnSO_4 + 5H_2O + CO_2$$

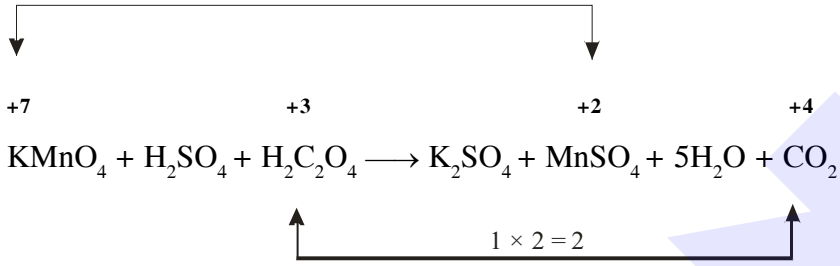
Sol. Step-I :

$$\overset{+7}{KMnO_4} + H_2SO_4 + \overset{+3}{H_2C_2O_4} \longrightarrow K_2SO_4 + \overset{+2}{MnSO_4} + 5H_2O + \overset{+4}{CO_2}$$

Multiply CO_2 by 2 to equalise the number of C-atoms

Step-II :

O.N. decreases by 5 per Mn atom



O.N. increases by 1 per C-atom

Step-III : Equalise increase and decrease in oxidation number by multiplying $KMnO_4$ by 2 and $H_2C_2O_4$ by 5.

$$2KMnO_4 + H_2SO_4 + 5H_2C_2O_4 \longrightarrow K_2SO_4 + 2MnSO_4 + H_2O + 10CO_2$$

Step-IV : Balance the atoms of other elements as

(a) K-atoms are already balanced

(b) S-atoms are three in right side and hence, it should also be three in left side.

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \longrightarrow K_2SO_4 + 2MnSO_4 + H_2O + 10CO_2$$

H-atoms are 16 in left side and hence it should also be 16 in right side.

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO$$

(d) Now check. There are 40 - O-atoms in both sides. Reaction is balanced.

6.2 Ion electron method or half cell method :

In this method, the two half equations are balanced separately and then added together to give balanced equation. The general procedure involves the following steps :

Step-I : Produce unbalanced equation for the reactions in ionic form.

Step-II : Separate the equation into two half reactions : oxidation half and reduction half.

Step-III : Balance the number of atoms of the elements by simple counting, for which changes in oxidation numbers are taking place.

Step-IV : For reactions occuring in acidic medium, add H_2O to balance O-atoms and H^+ to balance H-atoms.

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Step-V : Add electrons (s) to one side of half reactions to balance the charges. Make the numbers of electrons equal in both half reactions by multiplying them with suitable numbers.

Step-VI : Add both half reactions to achieve the overall reactions and cancel the electrons in both sides. Check the number of atoms of each element and total charges in each side, to reveal that the reaction is fully balanced in ionic form.

For the reaction in a basic medium, first balanced the atoms as is done in acidic medium. Then for each H^+ ion, add an equal number of OH^- ions to both sides of the equation. Where H^+ and OH^- appear on the same side of the equation , combine these to give H_2O.

Ex.6 Write the skeleton equation for each of the following processes and balance them by ion electron method :

(i) Permagnate ion oxidizes oxalate ions in acidic medium to carbon dioxide and gets reduced itself to Mn^{2+} ions.

(ii) Chlorine reacts with base to form chlorate ion and chloride ion.

Sol. (i) The skeleton equation for the process :

$$MnO_4^- + C_2O_4^{2-} + H^+ \longrightarrow Mn^{2+} + CO_2 + H_2O$$

$$\overset{+7}{Mn}O_4^- + \overset{+3}{C}_2O_4^{2-} + H^+ \longrightarrow Mn^{2+} + \overset{+4}{C}O_2 + H_2O$$

Step (I) : Writing oxidation and reduction half reaction :

$$\overset{+3}{C}_2O_4^{2-} \longrightarrow 2\overset{+4}{C}O_2 \qquad \text{(Oxidation half)}$$

$$\overset{+7}{Mn}O_4^- \longrightarrow Mn^{2+} \qquad \text{(Reduction half)}$$

Step (II) : Balancing 'O' and 'H'-atom by adding H_2O molecules and H^+ ions :

$$C_2O_4^{2-} \longrightarrow 2CO_2$$

$$MnO_4^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O$$

Step (III) : Balancing charge by electrons :

$$C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^-$$

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$

Step (IV) : Multiply the oxidation half reaction by 2 and reduction half reaction by 5 to equalize the electrons lost and gained and add the two half reactions.

$$[C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^-] \times 5$$

$$[MnO_4^- + 5e^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O] \times 2$$

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \longrightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$$

This is balance equation

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Sol. (ii) The skeleton equation for the given process :

$Cl_2 + OH^- \longrightarrow Cl^- + ClO_3^- + H_2O$

$\overset{0}{Cl_2} + OH^- \longrightarrow Cl^- + \overset{+5}{ClO_3^-} + H_2O$

Step (I) : Write the oxidation and reduction half reactions

$\overset{0}{Cl_2} \longrightarrow 2\overset{+5}{ClO_3^-}$ (Oxidation half)

$\overset{0}{Cl_2} \longrightarrow 2Cl^-$ (Reduction half)

Step (II) : Balancing O- and H-atom by H_2O molecules and H^+ ions.

$Cl_2 + 6H_2O \rightarrow 2ClO_3^- + 12\ H^+$

$Cl_2 \rightarrow 2Cl^-$

Step (III) : Balancing charge by electron(s)

$Cl_2 + 6H_2O \rightarrow 2ClO_3^- + 12\ H^+ + 10\ e^-$

$Cl_2 + 2\ e^- \rightarrow 2Cl^-$

Step (IV) : Multiply the reduction half reaction by 5 to make the number of electrons lost and gained equal and add both half reactions.

$Cl_2 + 6H_2O \rightarrow 2ClO_3^- + 12\ H^+ + 10\ e^-$

$[Cl_2 + 2\ e^- \rightarrow 2Cl^-\ ] \times 5$

---

or, $6Cl_2 + 6H_2O \rightarrow 2ClO_3^- + 12H^+ + 10Cl^-$

$3Cl_2 + 3H_2O \rightarrow ClO_3^- + 6H^+ + 5Cl$

Step (V) : As the reaction given is in basic medium, add 6 OH^- ions in each side.

$3Cl_2 + 3H_2O + 6OH^- \rightarrow ClO_3^- + 5Cl^- + 6H^+ + 6OH^-$

or $3Cl_2 + 3H_2O + 6OH^- \rightarrow ClO_3^- + 5Cl^- + 6H_2O$

or $3Cl_2 + 6OH^- \rightarrow ClO_3^- + 5Cl^- + 3H_2O$

This is balanced chemical equation.

7. Equivalent weight (E) :

7.1 Equivalent weight, $E = \frac{\text{Molecular weight}}{\text{valency factor (v.f)}}$

Valency factor may also be named as n-factor.

- Equivalent mass is a pure number. When expressed in gram, it is called gram equivalent mass.
- The equivalent mass of substance may have different values under different conditions, because n-factor depends on the nature of that substance in that reaction.

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7.2 n-factor calculation :

7.2.1 For Acids :

n-factor = number of replaceable H^+ ions = basicity of the acid

Here, basicity should be taken as moles of H^+ ions furnished per mole of the acid in water.

For example , for the acids HCl, H_2SO_4 and H_3PO_4, n-factors are 1,2 and 3 respectively and their equivalent weights are $\left(\frac{36.5}{1}=36\right)$, $\left(\frac{98}{2}=49\right)$ and $\left(\frac{98}{3}=31.67\right)$, respectively

(a) Non-metal oxides, which may also behave as acid in reactions :

(i) CO_2 :

$CO_2 + H_2O \rightarrow \underset{\text{carbonic acid}}{H_2CO_3} \rightarrow 2H^+ + CO_3^{2-}$

Hence, n-factor of $CO_2 = 2$

and Eq. wt. of $CO_2 = \frac{44}{2} = 22$

(ii) P_4O_{10} :

$P_4O_{10} + 6H_2O \rightarrow \underset{\text{Orthophosphoric acid}}{4H_3PO_4} \rightarrow 12H^+ + 4PO_4^{3-}$

Hence, n-factor of $P_4O_{10} = 12$

and Eq. wt. of $P_4O_{10} = \frac{284}{12} = 23.67$

(b) Partial neutralisation :

(i) $H_2SO_4 \rightarrow H^+ + HSO_4^-$

n-factor = 1

$E = \frac{98}{1} = 98$

(ii) $H_3PO_4 \rightarrow H^+ + H_2PO_4^-$

n-factor = 1

$E = \frac{98}{1} = 98$

(iii) $H_3PO_4 \rightarrow 2H^+ + HPO_4^{2-}$

n-factor = 2

$E = \frac{98}{2} = 49$

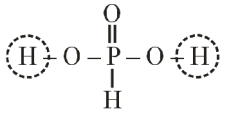
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(c) Special attention :

(i) H_3PO_3

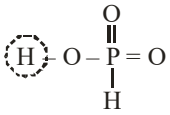
n-factor = 2

$$E = \frac{82}{2} = 41$$



(ii) H_3PO_2

n-factor = 1



$$E = \frac{66}{1} = 66$$

(iii) H_3BO_3

$B(OH)_3 + H_2O \rightarrow B(OH)_4^- + H^+$

Boric Acid

n-factor = 1

$$E = \frac{61.8}{1} = 61.8$$

7.2.2 For Base :

n-factor = number of replaceable OH^- ions = acidity of the base.

Here, acidity should be taken as moles of OH^- ions furnished per mole of base, in water, or moles of H^+ ions needed to neutralise each mole of the base.

For example, for the bases NaOH, $Ca(OH)_2$ and $Al(OH)_3$, the n-factors are 1,2 and 3, respectively and the equivalent weights are $\left(\frac{40}{1} = 40\right)$, $\left(\frac{74}{2} = 37\right)$ and $\left(\frac{78}{3} = 26\right)$ respectively.

(a) Metal oxides, which may also behave as base in reactions :

(i) Na_2O :

(i) $Na_2O + H_2O \rightarrow 2NaOH \rightarrow 2Na^+ + 2OH^-$

or $Na_2O + 2HCl \rightarrow 2NaCl + H_2O$

n-factor = 2

$$E = \frac{62}{2} = 31$$

(ii) Al_2O_3

$Al_2O_3 + 3H_2O \rightarrow 2Al(OH)_3 \rightarrow 2Al^{3+} + 6OH^-$

or $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$

n-factor = 6

$$E = \frac{102}{6} = 17$$

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(b) Partial neutralisation :

(i) $Mg(OH)_2 \rightarrow Mg(OH)^+ + OH^-$

n-factor = 1

$E = \frac{102}{6} = 58$

(ii) $Al(OH)_3 \rightarrow Al(OH)_2^+ + OH^-$

n-factor = 1

$E = \frac{78}{1} = 78$

(iii) $Al(OH)_3 \rightarrow Al(OH)^{2+} + 2OH^-$

n-factor = 2

$E = \frac{78}{2} = 39$

(c) N-containing base :

(i) NH_3

$NH_3 + HCl \rightarrow NH_4Cl$

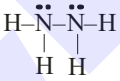


n-factor = 1

$E = \frac{17}{1} = 17$

(ii) N_2H_4 (hydrazine)

$N_2H_4 + 2HCl \rightarrow N_2H_6Cl_2$



n-factor = 2

$E = \frac{32}{2} = 16$

7.2.3 For Salts :

n-factor = Total number of positive charge or negative charge present in the salt.

For Example :

	Na_2CO_3	**$Fe_2(SO_4)_3$**	**$FeSO_4.7H_2O$**
	$(2Na^+ + CO_3^{2-})$	$(2Fe^{3+} + 3SO_4^{2-})$	$(Fe^{2+} + SO_4^{2-} + 7H_2O)$
n-factor	= 2	$2 \times 3 = 6$	2
E	$= \frac{M}{2}$	$\frac{M}{6}$	$\frac{M}{2}$

7.2.4 Eq. wt. of oxidising / reducing agents in redox reaction :

The equivalent weight of an oxidising agent is its weight which accepts or loose one mole electron in the chemical reaction.

The n-factor of oxidising or reducing is the number of electrons gained or lost per particle or change in oxidaiton number per particle

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(a) Eq. wt. of an oxidant or oxidising agent

$$= \frac{\text{Mol.wt.}}{\text{Mole of electrons gained per mole}} = \frac{\text{Mol.wt.}}{\text{decrease in oxidation no.}}$$

For example : $6e^- + Cr_2O_7^{2-} + 14H^+ \longrightarrow 2Cr^{3+} + 7H_2O$

Eq. wt. of $K_2Cr_2O_7 = \frac{\text{Mol. wt.}}{6}$

(b) Similarly eq. wt. of a reductant or reducing agent

$$= \frac{\text{Mol. wt.}}{\text{moles of electrons lost per mole}} = \frac{\text{Mol.wt.}}{\text{increase in oxidation no.}}$$

For example : $C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^-$

Eq. wt. of $H_2C_2O_4 = \frac{\text{Mol. wt.}}{2}$

(c) In different condition a compound may have different equivalent weights, because, it depends upon the number of electrons gained or lost by that compound in that reaction.

For example :

+7

(i) $MnO_4^- \longrightarrow Mn^{2+}$ (acidic medium)

Eq, wt. of $KMnO_4 = \frac{\text{Mol.wt.of } KMnO_4}{5} = \frac{158}{5} = 31.6$

(i) $\overset{+7}{M}nO_4^- \longrightarrow \overset{+4}{M}nO_2$ (neutral or basic medium)

Eq. wt. $KMnO_4 = \frac{\text{Mol.wt.of } KMnO_4}{3} = \frac{158}{3} = 52.7$

(iii) $\overset{+7}{M}nO_4 \longrightarrow \overset{+6}{M}nO_4^{2-}$ (strong alkaline medium)

Eq. wt. of $KMnO_4 = \frac{\text{Mol.wt.of } KMnO_4}{1} = 158$

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(d) **In any redox reaction, n-factor of all component is determined in terms of electrons n-actor is never determined as acid, base or salt, in redox reactions.** For example:

$$3\overset{0}{Cu} + 8H\overset{+5}{N}O_3 \rightarrow 3\overset{+2}{Cu}(NO_3)_2 + 3\overset{+2}{N}O + 4H_2O$$

The reaction is balanced by the loss or gain of 6 electrons per reaction. Hence,

n -factor or Cu = $\frac{6}{3} = 2$ and $E_{Cu} = \frac{M}{2}$

n -factor or $HNO_3 = \frac{6}{8} = \frac{3}{4}$ and $E_{HNO_3} = \frac{M}{3/4} = \frac{4M}{3}$

n -factor or $Cu(NO_3)_2 = \frac{6}{3} = 2$ and $E_{Cu(NO_3)_2} = \frac{M}{2}$

n -factor or NO = $\frac{6}{2} = 3$ and $E_{(NO)} = \frac{M}{3}$

and, n -factor or $H_2O = \frac{6}{4} = \frac{3}{2}$ and $E_{H_2O} = \frac{M}{3/2} = \frac{2M}{3}$

(e) **The n-factor of substance undergoing disporportionation (or comproportionation) is given**

as : n-factor = $\frac{n_1.n_2}{n_1+n_2}$

where n_1 and n_2 are n-factors of the substance only for oxidation and reductions half reactions. For example,

$$\overset{0}{Cl_2} \rightarrow Cl^- + \overset{+5}{ClO_3^-}$$

n_1 for Cl_2 undergoing oxidation only = $5 \times 2 = 10$

n_2 for Cl_2 undergoing reduciton only = $1 \times 2 = 2$

$\therefore$ n-facor of $Cl_2 = \frac{n_1.n_2}{n_1+n_2} = \frac{10\times 2}{10+2} = \frac{5}{3}$

and $E_{Cl_2} = \frac{M}{5/3} = \frac{3M}{5}$

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7.3 SOME OXIDIZING AGENTS WITH THEIR EQUIVALENT WEIGHTS :

Species	Changed to	Reaction	Electrons exchanged or change in O.N.	Eq. wt.
MnO_4^-	Mn^{+2} in acidic medium	$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$	5	$E = \frac{M}{5}$
MnO_4^-	MnO_2 in neutral medium	$MnO_4^- + 3e^- + 2H_2O \longrightarrow MnO_2 + 4OH^-$	3	$E = \frac{M}{3}$
MnO_4^-	MnO_4^{2-} in basic medium	$MnO_4^- + e^- \longrightarrow MnO_4^{2-}$	1	$E = \frac{M}{1}$
$Cr_2O_7^{2-}$	Cr^{3+} in acidic medium	$CrO_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$	6	$E = \frac{M}{6}$
MnO_2	Mn^{2+} in acidic medium	$MnO_2 + 4H^+ + 2e^- \longrightarrow Mn^{2+} + 2H_2O$	2	$E = \frac{M}{2}$
Cl_2	Cl^-	$Cl_2 + 2e^- \longrightarrow 2Cl^-$	2	$E = \frac{M}{2}$
H_2O_2	H_2O	$H_2O_2 + 2H^+ + 2e^- \longrightarrow 2H_2O$	2	$E = \frac{M}{2}$

7.4 SOME REDUCING AGENTS WITH THEIR EQUIVALENT WEIGHTS :

$S_2O_3^{2-}$	$S_4O_6^{2-}$	$2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + 2e^-$	2	$E = \frac{2M}{2}$
H_2O_2	O_2	$H_2O_2 \longrightarrow O_2 + 2H^+ + 2e^-$	2	$E = \frac{M}{2}$
Fe^{2+}	Fe^{3+}	$Fe^{2+} \longrightarrow Fe^{3+} + e^-$	1	$E = \frac{M}{1}$

8. g-EQUIVALENT OR EQUIVALENTS ($n_{eq.}$)

No. of g-equivalents , $n_{eq} = \frac{\text{wt.(in gm)}}{\text{gm eq.wt.}} = \frac{W}{E}$

Now, $E = \frac{M}{n - \text{factor}}$

$\therefore\ n_{eq.} = \frac{W}{M / n - \text{factor}} = \frac{W}{M} \times \text{n-factor} = n \times \text{n-factor}$

or, no. of g-equivalents = no. of moles × n-factors

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Ex.7 Calculate the number of g-equivalents in the following

(i) 4 gm NaOH

(ii) 2.45 gm H_2SO_4

(iii) 0.2 mole H_2O_2 (as O.A.)

(iv) 0.05 mole CH_4 (in complete combustion)

Sol. (i) n-factor of NaOH = 1

$$\therefore n_{eq} = n \times \text{n-factor} = \frac{4}{40} \times 1 = 0.1$$

(ii) n-factor of $H_2SO_4 = 2$

$$\therefore n_{eq} = n \times \text{n-factor} = \frac{2.45}{98} \times 2 = 0.05$$

(iii) $H_2\overset{-1}{O}_2 \xrightarrow{\text{As O.A.}} H_2\overset{-2}{O}$

n-factor of $H_2O_2 = 1 \times 2 = 2$

$\therefore n_{eq} = n \times n - \text{factor} = 0.2 \times 2 = 0.4$

(iv) $\overset{-4}{C}H_4 \longrightarrow \overset{+4}{C}O_2$

n-factor of $CH_4 = 8$

$\therefore n_{eq} = n \times \text{n-factor} = 0.05 \times 8 = 0.4$

Ex.8 Calculate the number of g-equivalents of solute in the following solution

(i) 100 ml 0.2 M - $KMnO_4$ (as O.A. in acidic medium)

(ii) 4L of 0.05 M - $HClO_3$ (in the change $ClO_3^- \rightarrow Cl^-$)

Sol. (i) $\overset{+7}{Mn}O_4^- \longrightarrow Mn^{2+}$

n-factor = 5

$$\therefore n_{eq.} = n \times \text{n-factor} = \left(\frac{100 \times 0.2}{1000}\right) \times 5 = 0.1$$

(ii) $\overset{+5}{Cl}O_3^- \longrightarrow Cl^-$

n-factor = 6

$\therefore n_{eq.} = n \times \text{n-factor} = (4 \times 0.05) \times 6 = 1.2$

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9. NORMALITY

Normality of solution is defined as the number of equivalent of solute present in one litre (1000 mL) solutions. Let a solution is prepared by dissolving W g of solute of eq. wt. E in V mL water.

- No. of equivalent of solute = $\frac{W}{E}$

$\because$ V mL of solution have $\frac{W}{E}$ equivalent of solute

$\therefore$ 1000 mL solution have $\frac{W \times 1000}{E \times V}$ equivalent of solute

$\therefore$ **Normality (N) =** $\frac{W \times 1000}{E \times V}$

- **Normality (N) = Molarity (M) × n-factor**

No. of g-equivalents of solute, $\mathbf{n_{eq}} = \frac{W}{E} = \frac{V.N}{1000}$

0.1 M - H_2SO_4 solution = 0.2 N - H_2SO_4 solution

Ex.9 4.9 gm H_3PO_4 is dissolved in sufficients water to get 300 ml solution. Calculate the normality of solution.

Sol. Eq. wt. of $H_3PO_4 = \frac{\text{Mol. wt.}}{\text{n – factor}} \Rightarrow E = \frac{98}{3}$

Now, normality = $\frac{W \times 1000}{E \times V} = \frac{4.9 \times 1000}{\frac{98}{3} \times 300} = 0.5$ N

Ex10. Calculate the no. of g-equivalents of the solute in 250 ml of 0.4 N-$KMnO_4$ solution.

Sol. $n_{eq} = \frac{V.N}{1000} = \frac{250 \times 0.4}{1000} = 0.1$

Ex11. The normality of $KMnO_4$ solution is 0.4 N w.r.t. reaction in acidic medium what should be the normality of same solution w.r.t. neutral medium.

Sol. Molarity of any solution is fixed but normality of same solution amy change, if n-factor of solute changes in different reactions, Now ,

M = N × n-factor = constant

or, 0.4 × 5 = N × 3

$\therefore$ Normality in neutral medium , N = $\frac{2}{3}$

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10. LAW OF EQUIVALENCE

The law states that one equivalent of a substance combine with one equivalent of the other substance and one equivalent of each product is formed. In a chemical reaction, equivalent (or mill equivalent) of reactants react in equal amount to give same no. of equivalent (or milli equivalents) of products separately.

$aA + bB \rightarrow mM + nN$

eq. of A = eq. of B = eq. of M = Eq. of N

or m.eq of A = m.eq of B = m.eq of M = m.eq of N

Ex.12 The number of moles of oxalate ions oxidized by one mole of MnO_4^- ion in acidic medium.

(A) $\frac{5}{2}$ ***(B)*** $\frac{2}{5}$ ***(C)*** $\frac{3}{5}$ ***(D)*** $\frac{5}{3}$

Sol : Equivalents of $C_2O_4^{2-}$ = equivalents of MnO_4^-

$x \text{ (mole)} \times 2 = 1 \times 5$

$x = \frac{5}{2}$

Ex.13 What weight of Na_2CO_3 of 85% purity would be required to react completely with 45.6 mL of 0.235N H_2SO_4 ?

Sol. Meq. of Na_2CO_3 = Meq. of H_2SO_4

or , $\frac{W_{Na_2CO_3}}{106/2} \times 1000 = 45.6 \times 0.235$

$\therefore \quad W_{Na_2CO_3} = 0.5679$ g

For 85 g of pure Na_2CO_3, weighed sample = 100 g

$\therefore$ For 0.5679 g of pure Na_2CO_3, weighed sample $= \frac{100}{85} \times 0.5679 = 0.6681$ g

Ex.14 ***Calculate the molar ratio in which the following two substances would react ? $Ba_3(PO_4)_2$ and $AlCl_3$***

Sol. $n_{eq}\ Ba_3(PO_4) = n_{eq}\ AlCl_3$

or $n_1 \times 6 = n_2 \times 3$

$\therefore\ n_1 : n_2 = 1 : 2$

Ex.15 Calculate volume of 0.4 M $KMnO_4$ required to react with following in acidic medium

$$KHC_2O_4(1 \text{ mol}) + H_2C_2O_4(2 \text{ mol})$$

Sol. $n_{eq}\ KMnO_4 = n_{eq}\ KHC_2O_4 + n_{eq}\ H_2C_2O_4$

or $(V \times 0.4) \times 5 = 1 \times 2 + 2 \times 2$

$\therefore\ V = 3.0$ L

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Ex.16. Calculate volume of 0.4 M NaOH required to react with following mixture.

$$KHC_2O_4(1 \text{ mole}) + H_2C_2O_4(2 \text{ mol})$$

Sol. n_{eq} NaOH = n_{eq} KHC_2O_4 + n_{eq} $H_2C_2O_4$

or , $(V \times 0.4) \times 1 = 1 \times 1 + 2 \times 2$

$\therefore$ V = 12.5 L

11. TITRATION OR TITRIMETRIC (VOLUMETRIC) ANALYSIS :

Volumetric analysis involves determination of the volume of a solution of accurately known concentration, which is required to react quantitatively with the measured volume of the solution of a substance, concentration of which is to be determined. The solution of accurately known concentration is called **standard solution**. The reagent of known concentration is called **titrant** and the substance being titrated is called **titrand**.

To carry out volumetric analysis, standard solution is usually added from the long graduated tube called **burette**.

The process of adding the standard solution to the solution of unknown concentration until the reactions in just complete is called **titration.**

11.1 REQUIREMENT FOR THE REACTION IN THE VOLUMETRIC ANALYSIS -

(i) The substance of which amount is to be determined by volumetric analysis must react completely and rapidly with the other reagent in stiochiometric proportion.

(ii) The reaction should be fast and their must be alternation in physical or chemical property of the solution at the equivalence poiont, which can be detected by an indicator, or by measuring the potential difference or current, etc.

11.2 STANDARD SOLUTION :

A solution of exactly known concentration is called standard solution. Any substance, which is stable at room temperature and does not react with solvent in which it is dissolved, can be directly weighed to prepared its standard solution. Description and preparation of the solutions is given below :

11.2.1 PRIMARY STANDARDS :

A primary standard is a compound of sufficient purity in which total amount of impurities does not exceed 0.01 - 0.02 %. The standard solution can be prepared by direct weighing of a sample of primary standard followed by its dissolution in water (or solvent) to obtained a defiite volume of solution. The substance to be used as a primary standard should also satisfy the following requirements:

(i) It must be easily avaialble in pure and dry form.

(ii) It should not undergo change in air. i.e. it should not be hydroscopic, oxidised by air or affected by gases such as carbon dioxide present in the atmosphere or lose water of crystallization, so that it can be stored safely.

(iii) It should be easy to detect the impurities present in it.

(iv) It should have high relative molecular mass so that weighing errors are neglible.

(vi) Its reaction with another substance shold be instantaneous and stoichiometric .

(vi) The substance should be readily soluble in water.

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It is difficult to obtain an ideal primary standard. Therefore, substances having characteristics neares to the prmary standars are usually employed.

Unstable hydrated salts, as a rule, should not be used as primary standards. However, sodium carbonate, sodium tetraborate, potassium hydrogenphthalate, oxalic acid, ferrous ammonium sulphate etc. can be used as primary standards because of their sufficients stabilities.

(a) Primary standary acids : Oxalic acid ($H_2C_2O_4.2H_2O$) , benzoic acid (C_6H_5COOH), potassium hydrogenphthalate ($KC_8H_5O_4$) ,etc.

(b) Primary standard bases : Washing soda (Na_2CO_3. 10 H_2O), borax ($Na_2B_4O_7.10H_2O$) , etc.

(c) Primary standard oxidising agent : $K_2Cr_2O_7$.

* $KMnO_4$ is not primary standard oxidsing agent because it undergo oxidation in water.

$4MnO_4^-$ (aq.) + $2H_2O(l) \rightarrow 4MnO_2(s) + 3O_2(g) + 4OH^-(aq.)$

(d) Primary standard reducing agent : Mohr's salt ($FeSO_4$. $(NH_4)_2SO_4.6H_2O$), Sodium oxalate ($Na_2C_2O_4$), etc.

11.2.2 SECONDARY STANDARDS :

A solution of secondary standary is the one which may be used for standardization after finding out its exact concentration by titration against a standard solution of primary standard.

A secondary standar cannot be used for preparing standard solution by direct weighing. Sodium hydroxide and potassium permanganate are examples of secondary standards.

11.3 EQUIVALENCE POINT :

The point at which reaction is completed in titration is called equivalence point or the theoretical or stoichimetric end point.

11.4 END POINT :

The point at which reaction of titration is stopped is called end point. The end point is detected either by some physical change produced in the reaction mixture itself or by addition of an auxiliary reagent, known as indicator ; alternatively some other physical measurements may be used. At the completion of the reaction, the indicator showns a visible change (i.e. , colour change or turbidity) in the solution being titrated.

In an ideal titration, the visible end point concides with the the equivalence point.

The role of indicator in colour change will be discussed in detail in the chapter, ionic equibrium.

11.5 Types of titration :

(a) ACID - BASE TITRATION :

(i) Acidimentry : Standard solution of acid is used to determine concentration of solution of base.

(ii) Alkalimetry : standard basic solution is used to determine the concentration of solution of acid.

(b) REDOX TITRATION :

(i) **Dichromometry :** Standard $K_2Cr_2O_7$ solution is used to determine the concentration of solution of reducing agent.

(ii) Permangnetometry : standard $KMnO_4$ solution is used to determine the concentraion of solution of reducing agent.

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(iii) Iodometry :

In iodometric titrations, an oxidizing agent is allowed to react in neutral medium or in acidic medium with excess of potassium iodide to liberate free iodine.

$$KI + \text{oxidizing agent} \longrightarrow I_2$$

Free iodine is titrated against a standard reducing agent usually with sodium thiosulphate. Halogen, dichromates, cupric ion, peroxides etc., can be estimated by this method.

$$I_2 + 2NaS_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$$

$$2CuSO_4 + 4KI \longrightarrow Cu_2I_2 + 2K_2SO_4 + I_2$$

$$K_4Cr_2O_7 + 6KI + 7H_2SO_4 \longrightarrow Cr_2(SO_4)_3 + 4K_2SO_4 + 7H_2O + 3I_2$$

❑ SOME IODOMETRIC TITRATIONS (TITRATING SOLUTIONS IS $Na_2S_2O_3.5H_2O$)

Estimation of	**Reaction**
I_2	$I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$
$CuSO_4$	$Cu^{2+} + 4I^- \longrightarrow Cu_2I_2 \downarrow + I_2$
$CaOCl_2$	$Cl_2 + 2I^- \longrightarrow 2Cl^- + I_2$
MnO_2	$MnO_2 + 4H^+ + 2Cl^- \longrightarrow Mn^{2+} + 2H_2O + Cl_2$
	$Cl_2 + 2I^- \longrightarrow I_2 + 2Cl^-$
IO_3^-	$IO_3^- + 5I^- + 6H^+ \longrightarrow 3I_2 + 3H_2O$
H_2O_2	$H_2O_2 + 2I^- + 2H^+ \longrightarrow I_2 + 2H_2O$
Cl_2	$Cl_2 + 2I^- \longrightarrow 2CI^- + I_2$
O_3	$O_3 + 2I^- + 2H^+ \longrightarrow I_2 + H_2O + O_2$
$Cr_2O_7^{2-}$	$Cr_2O_7^{2-} + 14H^+ + 6I^- \longrightarrow 3I_2 + 2Cr^{3+} + 7H_2O$
MnO_4^-	$2MnO_4^- + 10I^- + 16H^+ \longrightarrow 2Mn^{2+} + 5I_2 + 8H_2O$

(iv) Iodimetry :

Stadnard I_2 solution is used to determine the concentration of solution of reducing agent.

❑ SOME IODIMETRIC TITRATION (TITRATING SOLUTIONS IS I_2 IN KI)

Estimation of	**Reaction**
H_2S (in acidic medium)	$H_2S + I_2 \longrightarrow S + 2I^- + 2H^+$
SO_3^{2-} (in acidic medium)	$SO_3^{2-} + I_2 + H_2O \longrightarrow SO_4^{2-} + 2I^- + 2H^+$
Sn^{2+} (in acidic medium)	$Sn^{2+} + I_2 \longrightarrow Sn^{4+} + 2I^-$

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In iodimetric and iodometric titration, starch solution is used as an indicator. Starch solution gives blue or violet colour with free iodine. At the end point, the blue or violet colour disappears when iodine is completely changed to iodide.

11.6 Calculation of concentration :

Let there be a solution of substance A of unknown concentration. We are given solution of another substance B whose concentration is known (N_1). We take a certain known volume of A in a flask (V_2) and then we add B to A slowly till all the A is consumed by B (this can be known with the help of indicators). Let us, assume that the volume of B consumed is V_1. According to the law of equivalence, the number of g equivalents of B at the end point.

$\therefore N_1V_1 = N_2V_2$,

where N_2 is the conc. of A. From this we can calculate the value of N_2.

11.7 BACK TITRATION

Back titration is a titration done in reverse ; instead of titrating the original sample, a known excess of standard reagent is added to the solution, and the excess is titrated. A back titration is useful if the endpoint of the reverse titration is easier to identify than the endpoint of the normal titration, as with the precipitation reactions.

Ex.17 How many milliliters of 0.02000 M $KMnO_4$ solution would be required to exactly titrate 25.00 mL of 0.2000 M $Fe(NO_3)_2$ solution.

Sol : $MnO_4^- + Fe^{2+} \rightarrow Mn^{2+} + Fe^{3+}$

n-factor = 5 n-factor = 1

Meq of $KMnO_4$ = Meq. of $Fe(NO_3)_2$

or, $V \times (0.02 \times 5) = 25 \times (0.2 \times 1)$

$\therefore V = 50$ ml

Ex.18 3.55 g sample of bleaching powder suspended in H_2O was treated with enough acetic acid and KI solution. Iodine thus liberated required 80 mL of 0.2 M hypo for titration. Calculate the % of available chlorine.

Sol. n_{eq} available chlorine = n_{eq} I_2 = n_{eq} hypo

$$\text{or, } \frac{w}{35.5} = \left(\frac{80 \times 0.2}{1000}\right) \times 1 \qquad [\text{hypo is } Na_2S_2O_3.\,5H_2O]$$

$\therefore w = 16 \times 35.5 \times 10^{-3}$ gm

$$\text{Hence, \% of available chlorine} = \frac{16 \times 35.5 \times 10^{-3}}{3.55} \times 100 = 16\%$$

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Ex.19 50 gm of a sample of $Ca(OH)_2$ is dissolved in 50 ml of 0.5 N HCl solution. The excess of HCl was titrated with 0.3N – NaOH. The volume of NaOH used was 20cc. Calculate % purity of $Ca(OH)_2$

Sol. $n_{eq.}\ Ca(OH)_2 + n_{eq.}\ NaOH = n_{eq.}\ HCl$

$$\text{or, } \left(\frac{w}{74}\right) \times 2 + \left(\frac{20\times 0.3}{1000}\right) = \frac{50\times 0.5}{1000}$$

$\therefore$ wt. of $Ca(OH)_2$ taken, w = 0.703 gm

$$\text{Hence, \% purity} = \frac{0.703}{50}\times 100 = 1.406\ \%.$$

Ex.20 To 50 L of 0.2 N NaOH, 2.5 L of 2N HCl and 15 L of 0.1 N $FeCl_3$ solutions are added. What weight of Fe_2O_3 can be obtained from the precipitate? Also report the normality of NaOH left in resultant solution :

Sol. $n_{eq.}$ NaOH taken = 50 × 0.2 = 10

$n_{eq.}$ HCl taken = 2.5 × 2 = 5

$n_{eq.}\ FeCl_3$ taken = 15 × 0.1 = 1.5

NaOH will react with both , HCl and $FeCl_3$, and from the amounts taken, its is clear that NaOH is in excess.

Now, $n_{eq.}\ FeCl_3 = n_{eq.}\ Fe_2O_3$ formed

$$\text{or , } 1.5 = \left(\frac{W}{160}\right)\times 6$$

$\therefore$ wt of Fe_2O_3 formed , w = 40 gm.

Now, $n_{eq.}$ NaOH remained = 10 – (5 + 1.5) = 3.5 and the final volume of solution

= 50 + 2.5 + 15 = 67.5 L

$$\therefore \text{ Normality of NaOH left} = \frac{3.5}{67.5} = 0.052\ N$$

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Ex.21 ***1.20 g sample of Na_2CO_3 and K_2CO_3 was dissolved in water to form 100 mL of a Solution. 20 mL of this solution required 40 mL of 0.1 N HCl for complete neutralization. Calculate the weight of Na_2CO_3 in the mixture. If another 20 mL of this solution is treated with excess of $BaCl_2$ what will be the weight of the precipitate ?***

Sol. Let, weight of Na_2CO_3 = x g

Weight of K_2CO_3 = y g

$\therefore$ x + y = 1.20 g .....(i)

For neutralization reaction of 100 mL

Meq. of Na_2CO_3 + Meq. of K_2CO_3 = Meq. of HCl

$$\Rightarrow \frac{x}{106} \times 2 \times 1000 + \frac{y}{138} \times 2 \times 1000 = \frac{40 \times 0.1 \times 100}{20}$$

$\therefore$ 69 x + 53 y = 73.14 .....(ii)

From Eqs. (i) and (ii), we get

x = 0.5962 g

y = 0.604 g

Solution of Na_2CO_3 and K_2CO_3 gives ppt. of $BaCO_3$ with $BaCl_2$

(Meq. of Na_2CO_3 + Meq. of K_2CO_3) in 20 mL = Meq. of $BaCO_3$

$\Rightarrow$ Meq. of HCl for 20 mL mixture = Meq. of $BaCO_3$

$\Rightarrow$ Meq. of $BaCO_3$ = 40 × 0.1 = 4

$$\frac{W_{BaCO_3}}{M_{BaCO_3}} \times 1000 = 40 \times 0.1 = 4$$

$$\frac{W_{BaCO_3}}{197} \times 2 \times 1000 = 4$$

$\therefore$ W_{BaCO_3} = 0.394 g

Ex.22 One gm of impure sodium carbonate is dissolved in water and the solution is made up to 250 ml. To 50 ml of this made up solution, 50 ml of 0.1N – HCl is added and the mix after shaking well required 10 ml of 0.16 N – NaOH solution for complete titration. Calculate the % purity of the sample.

Sol. $n_{eq.}$ Na_2CO_3 (in diluted solution) + n_{eq} NaOH. = $n_{eq.}$ HCl

$$\text{or } \left(\frac{50 \times M}{1000}\right) \times 2 + \frac{10 \times 0.16}{1000} = \frac{50 \times 0.1}{1000}$$

$\therefore$ Molarity of diluted solution, M = 0.034

$$\text{Hence, moles of } Na_2CO_3 \text{ taken} = \frac{250 \times 0.034}{1000} = 8.5 \times 10^{-3}$$

$$\therefore \text{ \% purity of sample} = \frac{8.5 \times 10^{-3} \times 106}{1} \times 100 = 90.1\%$$

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Ex.23 Mg can reduce NO_3^- to NH_3 in basic medium.

$NO_3^- + Mg\ (s) + H_2O \rightarrow Mg\ (OH)_2\ (s) + OH^-\ (aq.) + NH_3(g)$

A 25.0 mL sample of NO_3^- solution was treated with Mg. The NH_3 (g) was passed into 100 mL of 0.15 N HCl. The excess of HCl required 32.10 mL of 0.10 N NaOH for neutralization. What was the molarity of NO_3^- ions in the original sample ?

Sol. From conservation of N-atoms,

moles of NO_3^- = moles of NH_3 = n

Now, $n_{eq.}$ HCl = $n_{eq.}$ NH_3 + $n_{eq.}$ NaOH

or, $\frac{100\times0.15}{1000} = n \times 1 + \frac{32.10\times0.10}{1000}$

$\therefore$ moles of NO_3^-, $n = 11.79 \times 10^{-3}$

Now, $\frac{25\times M_{NO_3^-}}{1000} = 11.79\times10^{-3}$

$\therefore$ molarity of original NO_3^- solution, $M_{NO_3^-} = 0.47M$

Ex.24 3.2 g of pyrolusite (MnO_2) was treated with 50 mL of 0.5 M oxalic acid and some sulphuric acid. The oxalic acid left undecomposed was raised to 250 mL in a flask. 25 mL of this solution when treated with 0.02 M $KMnO_4$ required 32 mL of the solution : Find the % of MnO_2 in the sample.

Sol. $\overset{+4}{Mn}O_2 + H_2\overset{+3}{C_2}O_4 \xrightarrow{H^+} \overset{2+}{Mn} + \overset{2+}{CO_2}$

(n-factor = 2) (n-factor = 2)

$H_2\overset{+3}{C_2}O_4 + \overset{+7}{Mn}O_4^- \xrightarrow{H^+} \overset{2+}{CO_2} + \overset{2+}{Mn}$

(n-factor = 2) (n-factor = 5)

$H_2C_2O_4$ reacts with MnO_2 and the remaining $H_2C_2O_4$ reacts with $KMnO_4$.

Moles of $H_2C_2O_4$ taken = $\frac{50\times0.5}{1000} = 25 \times 10^{-3}$

Now, $n_{eq.}$ $H_2C_2O_4$ = n_{eq} $KMnO_4$

or, $n \times 2 = \left(\frac{32\times0.02}{1000}\right)\times5$

$\therefore$ moles of $H_2C_2O_4$ remained in 25 ml solution, $n = 1.6 \times 10^{-3}$.

Hence, moles of $H_2C_2O_4$ with $MnO_2 = 25 \times 10^{-3} - 1.6 \times 10^{-3} \times \frac{250}{25} = 9\times10^{-3}$

Now, neq. MnO_2 = neq $H_2C_2O_4$

or, $\left(\frac{w}{87}\right)\times2 = 9 \times 10^{-3} \times 2$

or, $w = 9 \times 87 \times 10^{-3}$ gm

$\therefore$ % of pure MnO_2 in pyrolusite = $\frac{9\times87\times10^{-3}}{3.2} \times 100 = 24.47\ \%$

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Ex.25 An aqueous solution containing 0.10 g KIO_3 (formula wt. 214.0) was treated with an excess of KI solution. The solution was acidified with HCl. The liberated I_2 consumed 45.0 mL of thiosulphate solution to decolourise the blue starch – iodine complex. Calculate the molarity of the sodium thiosulphate solution. ***[JEE 1998]***

Sol. $KIO_3 + 5KI + 6HCl \rightarrow 6KCl + 3I_2 + 3H_2O$

$\frac{0.1}{214}$mole $\qquad\qquad$ $3\times\frac{0.1}{214}$mole

$I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

$\frac{0.3}{214}$ mole $\quad 2\times\frac{0.3}{214}$mole $= \frac{45\times M}{1000}$

$\therefore$ molarity of $Na_2S_2O_3$ solution = 0.062 M

12. HARDNESS OF WATER :

Presence of calcium and magnesium salts in water makes water, hard. **Hard water** does not give lather with soap.

Water free from soluble salts of calcium and magnesium is called **Soft water**. It gives lather with soap easily.

Hard water forms scum/precipitate with soap. Soap containing sodium stearate ($C_{17}H_{35}COONa$) reacts with hard water to precipitate out Ca/Mg stearate.

$2C_{17}H_{35}COONa(aq) + M^{2+}(aq) \rightarrow (C_{17}H_{35}COO)_2M\downarrow + 2Na^+(aq)$; M is Ca/Mg

Hardness of water is of two types

(a) Temporary Hardness

(b) Permanent Hardness

12.1 TEMPORARY HARDNESS :-

This is due to presence of bicarbonate of calcium and magnesium. Rain water dissolves a small quantities of CO_2 from the atmospher forming a very dilute solution of carbonic acid.

This water attacks Ca and Mg carbonate in any rock

$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$

Temporary hardness in water is easily removed by boiling, as the bicarbonates decomposes readily and the insoluble carbonates are precipitated.

$Ca(HCO_3)_2 \xrightarrow{\text{Boil}} CaCO_3 + H_2O + CO_2$ (Insoluble)

$Mg(HCO_3)_2 \xrightarrow{\text{Boil}} Mg(OH)_2 + 2CO_2$ (Insoluble)

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Temporarily hardness can also be removed by **clark's process** which involves the addition of slaked lime Ca(OH)

$$Ca(HCO_3)_2 + Ca(OH)_2 \xrightarrow{\text{Boil}} 2CaCO_3 + 2H_2O$$

(Insoluble)

$$Mg(HCO_3)_2 + 2Ca(OH)_2 \longrightarrow Mg(OH)_2 + 2CaCO_3 + 2H_2O$$

(Insoluble) (Insoluble)

12.2 PERMANENT HARDNESS :

Permanent hardness is due to presence of sulphates and chlorides of both calcium and magnesium. This type of hardness cannot be removed by boiling or by $Ca(OH)_2$; Substance used to remove the hardness of water are known as water softener. This various water softeners are -

(i) **Washing soda :-** It removes both the temporary and permanent hardness by converting soluble Ca and Mg compounds into insoluble carbonates.

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 \downarrow + 2NaCl$$

$$CaSO_4 + Na_2CO_3 \longrightarrow CaCO_3 \downarrow + Na_2SO_4$$

(ii) **Permutit:** Permutit is technical name given to certain hydrated silicates of aluminium and sodium.or sodium alumunium orthosilicate or sodium zeolite. This method is also called ion - exchange method or zeolite method.

Example : $Na_2O \cdot Al_2O_3 \cdot 2SiO_2 \cdot xH_2O$ (symbolised as Na_2Z)

$$Na_2Z + Ca^{+2} \longrightarrow CaZ + 2Na^{\oplus}$$

$$Na_2Z + Mg^{+2} \longrightarrow MgZ + 2Na^{\oplus}$$

These ions can be re-exchanged by treating it with brine (NaCl) solution.

$$CaZ + 2NaCl \longrightarrow Na_2Z + CaCl_2$$

This method is useful for the removal of both temporary and permanent hardness of water.

(iii) **Calgon :** The complex salt of metaphosphoric acid, sodium hexametaphosphate $(Na_4(PO_3)_6$ is called Calgon. It is represented as $Na_2[Na_4(PO_3)_6]$.

$$2CaSO_4 + Na_2[Na_4(PO_3)_6] \longrightarrow Na_2[Ca_2(PO_3)_6] + Na_2SO_4$$

$$2MgSO_4 + Na_2[Na_4(PO_3)_6] \longrightarrow Na_2[Mg_2(PO_3)_6] + Na_2SO_4$$

The complex formed keep Ca and Mg ions in water but unreactive towards soaps.

(iv) **Ion exchange Resins:-** Ion exchange resins are the most popular water softener thease days. This resins are synthetic substance. The cation exchanger consists of granular insoluble organic acid resins having giant molecules with $-SO_3H$ or -COOH groups while the anion exchanges contains giant organic molecules with basic groups derived from amine. Ion exchange resins removes all soluble mineral from water.

Hard water $\longrightarrow$ [ ] $\longrightarrow$ [ ] $\longrightarrow$ Pure water

Cation exchanger	Anion exchanger
(Remove cation like Na^+, Mg^{2+} or Ca^{2+} and furnish H^+)	(Remove anion like SO_4^{2-}, Cl^- or NO_3^- and furnish OH^-)

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The water coming from cation exchanger is acidic due to H^+ ions This water is then passed through another bed containing anion exchanger. This exchanger removes anion like Cl^-, SO_4^{2-}, NO_3^- by exchanging with OH^- ions.

Reaction at Cation exchanger :

$$2RH(s) + Ca^{2+}(aq.) \longrightarrow R_2Ca(s) + 2H^+(aq.)$$

$$RH(s) + Na^+(aq.) \longrightarrow RNa(s) + H^+(aq.)$$

Reaction at Anion exchanger :

$$RNH_2(s) + H_2O(l) \longrightarrow RNH_3^+.OH^- (s)$$

$$RNH_3^+.OH^-(s) + Cl^-(aq.) \longrightarrow RNH_3^+.Cl^- (s) + OH^-(aq.)$$

$$RNH_3^+.OH^-(s) + SO_4^{2-}(aq.) \longrightarrow (RNH_3^+)_2.SO_4(s) + 2OH^-(aq.)$$

OH^- produced neutralises H^+ ions in water from cation exchange.

$$H^+(aq.) + OH^-(aq.) \longrightarrow H_2O(l)$$

The exhausted cation and anion exchange resin beds are regenerated by treatment with dilute acid and akali solutions, respectively.

12.3 DEGREE OF HARDNESS :

Degree of Hardness of water is measured in terms of ppm of $CaCO_3$ which is defined as number of parts of $CaCO_3$ by mass present in one million parts by mass of water, equivalent to the of various calcium & magnesium salts.

For example, consider the sample of hard water which is found to contain 36 mg of $MgSO_4$ per kg of water.

gm eq. of $CaCO_3$ = gm eq. of $MgSO_4$

$$\frac{W_{CaCO_3}}{100} \times 2 = \frac{36 \times 10^{-3}}{120} \times 2$$

$$W_{CaCO_3} = 3 \times 10^{-2} gm$$

$$\text{degree of hardness} = \frac{3 \times 10^{-2}}{1000} \times 10^6 = 30 \text{ ppm } CaCO_3$$

Ex.26 50 L sample of tap water contains 2.40 gm $MgSO_4$ and 1.17 gm NaCl. Calculate the hardness of water.

Sol. Hardness is due to $MgSO_4$ only

$$n_{eq.}\ CaCO_3 = n_{eq.}\ MgSO_4$$

$$\text{or, } \left(\frac{w}{100}\right) \times 2 = \left(\frac{2.40}{120}\right) \times 2$$

$$\therefore w_{CaCO_3} = 2gm$$

$$\text{Hence, hardness} = \frac{2}{50 \times 10^3} \times 10^6 = 40 \text{ ppm } CaCO_3$$

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Ex.27 200 ml of a water sample contain 8.2 mg $Ca(NO_3)_2$, 5.7 $MgCl_2$, 6.48 mg $Ca(HCO_3)_2$ and 2.13 mg Na_2SO_4. Calculate the temporary, permanent and total hardness of water sample.

Sol. Temporary hardness due to $Ca(HCO_3)_2$ only

$n_{eq.}\ CaCO_3 = n_{eq.}\ Ca(HCO_3)_2$

or, $\left(\frac{w}{100}\right)\times 2 = \left(\frac{6.48}{162}\right)\times 2$

$\therefore\ w_{CaCO_3} = 4$ mg

Hence, temporary hardness = $\frac{4\times10^{-3}}{200}\times10^{6} = 20$ ppm $CaCO_3$

Permanent hardness is due to $Ca(NO_3)_2$ and $MgCl_2$ only

$n_{eq.}\ CaCO_3 = n_{eq.}\ Ca(NO_3)_2 + n_{eq.}\ MgCl_2$

or, $\left(\frac{w}{100}\right)\times 2 = \left(\frac{8.2}{164}\right)\times 2 + \left(\frac{5.7}{95}\right)\times 2$

$\therefore\ w_{CaCO_3} = 11$ mg

Hence, permanent hardness = $\frac{11\times10^{-3}}{200}\times10^{6} = 55$ ppm $CaCO_3$

and , the total hardness = 20 + 55 = 75 ppm. $CaCO_3$

Ex.28 50 ml of tap water exactly require 20 ml (N/50) HCl solution for titration of all the bicarbonates present in water. Calculate the temporary hardness in water.

Sol. $n_{eq.}\ HCl = n_{eq.}\ HCO_3^- = n_{eq.}\ CaCO_3$

or, $\frac{20\times\frac{1}{50}}{1000} = \left(\frac{w}{100}\right)\times 2$

$\therefore\ w_{CaCO_3} = 0.02$ mg

$\therefore$ Temporary hardness = $\frac{0.02}{50}\times10^{6} = 400$ ppm $CaCO_3$

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Ex.29 100 ml sample of hard water containing only Ca^{2+} hardness is passed through a column of cation exchange resin (H^+ – resin). The water coming out the column require 20 ml of 0.02M NaOH for its titration. What is hardness of water as ppm of Ca^{2+}

Sol. $n_{eq.}$ NaOH = $n_{eq.}$ H^+ = $n_{eq.}$ Ca^{2+}

or, $\left(\frac{20\times0.02}{1000}\right)\times1 = \left(\frac{w}{40}\right)\times2$

$\therefore$ $w_{Ca^{2+}} = 8 \times 10^{-3}$ gm

Hence, hardness = $\frac{8\times10^{-3}}{100} \times 10^6 = 80$ ppm Ca^{2+}

Ex.30 When 50 gm hard water containing 40 ppm Ca^{2+} ions and 54 ppm Al^{3+} ions is passed through cation exchange resin (acidic resin), the final water sample requires 40 ml NaOH solution for titration. Calculate the normality of NaOH sample used.

Sol. Mass of Ca^{2+} ions in hard water = $\frac{40}{10^6}\times50 = 2 \times 10^{-3}$ gm

and mass of Al^{3+} ions in hard water = $\frac{54}{10^6}\times50 = 2.7 \times 10^{-3}$ gm

Hence, moles of H^+ produced = $\left(\frac{2\times10^{-3}}{40}\right)\times2 + \left(\frac{2.7\times10^{-3}}{27}\right)\times3 = 4 \times 10^{-4}$

Now. $n_{eq.}$ H^+ = $n_{eq.}$ OH^-

or, $4 \times 10^{-4} \times 1 = \frac{40\times N_{OH^-}}{1000}$

$\therefore$ Normality of NaOH solution = 0.01 N

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SOME MORE SOLVED EXAMPLES

Ex.31 A 3.00 g sample containing Fe_3O_4 , Fe_2O_3 and an inert impure substance, is treated with excess of KI solution in presence of dilute H_2SO_4. The entire iron is converted into Fe^{2+} along with the liberation of iodine. The resulting solution is diluted to 100 mL . A 20 mL of the diluted solution require 11 mL of 0.5 M $Na_2S_2O_3$ solution to reduce the iodine present. A 50 mL of diluted solution after complete extraction of the iodine requires 12.80 mL of 0.25 $KMnO_4$ solution in dilute H_2SO_4 medium for the oxidation of Fe^{2+}. Calculate the percentages of Fe_2O_3 and Fe_3O_4 in the original sample.

Sol. Let number of moles of Fe_3O_4 is 'a' and no. of moles of Fe_2O_3 is 'b' in 3 gram sample.

Now, 'a' moles Fe_3O_4 means (a moles Fe_2O_3 + a mole of FeO)

So, in sample (a + b) moles Fe_2O_3 and 'a' mole FeO present.

$Fe_2O_3 + KI + \xrightarrow{H_2SO_4} FeSO_4 + I_2 + K_2SO_4 + H_2O$

No. of gram equivalent of Fe_2O_3 = no. of gram equivalent of I_2

$n_{Fe_2O_3} \times n_{f(Fe_2O_3)} = n_{I_2} \times n_{f(I_2)}$

$(a + b) \times 2 = n_{I_2} \times 2$

$n_{I_2} = (a + b)$ moles.

Now, solution is diluted to 100 ml

$$\text{So, } [I_2] = \frac{n}{V_{litre}} = \frac{(a+b)}{100/1000} = 10(a+b)$$

$$[Fe^{2+}] = \left(\frac{n(Fe^{2+})}{V_{litre}}\right) = \frac{(3a+2b)}{100/1000} = 10(3a + 2b)$$

Now 20 ml of this solution titrated with 11 ml, 0.5 M $Na_2S_2O_3$

$I_2 + Na_2S_2O_3 \rightarrow NaI + Na_2S_4O_6$

No. of gram equivalent of I_2 = no. of gram equivalent of $Na_2S_2O_3$

$(N \times V)_{I_2} = (N \times V)_{Na_2S_2O_3}$

$(M \times n_f \times V)_{I_2} = (M \times n_f \times V)_{Na_2S_2O_3}$

$10 (a + b) \times 2 \times 20 = 0.5 \times 1 \times 11$

$$a + b = \frac{11}{800} \quad \text{..... (i)}$$

50 ml of solution after extraction of I_2 allowed to oxidation of Fe^{2+} by 12.80 ml, 0.25 M $KMnO_4$.

$Fe^{2+} + KMnO_4 + H_2SO_4 \rightarrow Fe^{3+} + MnSO_4 + K_2SO_4 + H_2O$

no. of gram equivalent of Fe^{2+} = no. of gram equivalent of $KMnO_4$.

$(M \times n_f \times V)_{Fe^{2+}} = (M \times n_f \times V)_{KMnO_4}$

$10(3a + 2b) \times 1 \times 50 = 0.25 \times 5 \times 12.80$

$$3a + 2b = \frac{16}{500} \quad \text{.....(ii)}$$

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By (i) & (ii)

a = 0.0045 moles

b = 0.00925 moles

$$\text{so } \%\ Fe_2O_3 = \frac{Wt(Fe_2O_3)}{\text{Total wt.}} \times 100 = \frac{b\times160}{3}\times100 = \frac{0.00925\times160}{3}\times100 = 49.33\%.$$

$$\%\ Fe_2O_3 = \frac{Wt(Fe_2O_3)}{\text{Total wt.}} \times 100 = \frac{a\times232}{3}\times100 = \frac{0.0045\times232}{3}\times100 = 34.8\ \%$$

Ex.32 Hydrogen peroxide solution (20 mL) reacts quantitatively with a solution of $KMnO_4$ (20 mL) acidified with dilute H_2SO_4. The same volume of $KMnO_4$ solution is just decolorized by 10 mL of $MnSO_4$ in neutral medium simultaneously forming a dark brown precipitate of hydrated MnO_2. The brown precipitate is dissolved in 10 mL of 0.2 M sodium oxalate under boiling condition in the presence of dilute H_2SO_4. Write the equations involved in the reactions and calculate the molarity of H_2O_2.

Sol. Let molarity of H_2O_2 solution is M_1

20 ml, M_1 molar H_2O_2 react with 20 ml $KMnO_4$ in H_2SO_4 solution

$$H_2O_2 + KMnO_4 + H_2SO_4 \rightarrow MnSO_4 + K_2SO_4 + H_2O + O_2$$

no. of gram equivalent of H_2O_2 = no. of gram equivalent of $KMnO_4$

$$(M \times n_f \times V)_{H_2O_2} = (M \times n_f \times V)_{KMnO_4}$$

$$M_1 \times 2 \times 20 = M_{KMnO_4}\times5\times20 \qquad \text{.....(i)}$$

20 ml $KMnO_4$ decolourise by 10 ml $MnSO_4$ in neutral medium.

$KMnO_4 + MnSO_4 + H_2O \rightarrow MnO_2 + H_2SO_4 + KHSO_4$

no. of gram equivalent of $KMnO_4$ = no. of gram equivalent of MnO_2

$M_{KMnO_4}\times3\times20\text{ml}$ = no. of gram equivalent of MnO_2

$$M_{KMnO_4}\times3\times20\times10^{-3} = n_{MnO_2}\times6/5 \qquad \text{.....(ii)}$$

and obtained MnO_2 dissolved in 10 ml, 0.2 M $Na_2C_2O_4$

$MnO_2 + Na_2C_2O_4 + H_2SO_4 \rightarrow MnSO_4 + CO_2 + Na_2SO_4 + H_2O$

no. of gram equivalent of MnO_2 = no. of gram equivalent of $Na_2C_2O_4$

$$n_{MnO_2}\times2 = 0.2\times2\times10\times10^{-3}$$

$$n_{MnO_2} = 2\times10^{-3}\text{ moles} \qquad \text{.....(iii)}$$

From equaiton (ii)

$$M_{KMnO_4} \times 3 \times 20 \times 10^{-3} = 2 \times 10^{-3} \times \frac{6}{5}$$

$$M_{KMnO_4} = \frac{2}{50}$$

From equation (1)

$$M_1 \times 2 \times 20 = \frac{2}{50}\times5\times20$$

$$M_1 = 0.1M$$

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Ex.33 One litre of a mixture of O_2 and O_3 ($O_3 \longrightarrow O_2 + O^{2-}$) at 1 atm and 273 K was allowed to react with an excess of acidified solution of KI. The iodine liberated required 40 ml of M/10 sodium thiosulphate solution for titration. What is the percent of ozone in the mixture ?

Sol. Let in 1 litre mixture x litre is O_2 and y litre is O_3 at NTP

excess KI reduce O_3 into O_2 and O^{2-}

$KI + O_3 + H_2O \rightarrow I_2 + O_2 + KOH$

no. of gram equivalent of O_3 = no. of gram equivalent of I_2

$(n \times n_f)_{O_3} = (n \times n_f)_{I_2}$

$$\frac{y}{22.4} \times 2 = n_{I_2} \times 2$$

$$n_{I_2} = \frac{y}{22.4} \qquad .....(1)$$

Now, released I_2 required 40 ml , $\frac{1}{10}$M $Na_2S_2O_3$

$I_2 + Na_2S_2O_3 \rightarrow NaI + Na_2S_4O_6$

No. of gram equivalent of I_2 = no. of gram equivalent of $Na_2S_2O_3$

$$n_{I_2} \times 2 = \frac{1}{10} \times 1 \times 40 \times 10^{-3}$$

$n_{I_2} = 2 \times 10^{-3}$ moles

Now from equation (1)

$$\frac{y}{22.4} = 2 \times 10^{-3}$$

$y = (44.8 \times 10^{-3})$ litre $= 0.0448$ litre

$x = 1 - y = 1 - 0.0448 = 0.9552$ litre

$$\% \text{ (by wt)} = \frac{wt(O_3)}{wt.(total)} \times 100 = \frac{n(O_3) \times 48}{n(O_3) \times 48 + n(O_2) \times 32} \times 100$$

$$= \frac{\frac{y}{22.4} \times 48}{\frac{y}{22.4} \times 48 + \frac{x}{22.4} \times 32} \times 100 = \frac{3y}{3y + 2x} \times 100$$

$$= \frac{3 \times 0.0448}{3 \times 0.048 + 2 \times 0.9552} \times 100 = 6.57\ \%$$

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Ex.34 A mixture of FeO and Fe_2O_3 is reacted with acidified $KMnO_4$ solution having a concentration of 2/5 M, 100 mL of which was used. The solution was then titrated with Zn dust which converted Fe^{3+} of the solution to Fe^{2+}. The Fe^{2+} required 1000 mL of 2/15 M $K_2Cr_2O_7$ solution. Find the % by mol of FeO and Fe_2O_3.

Sol. Let no. of moles of FeO is x millimoles & Fe_2O_3 is y millimoles in mixture.

$FeO + KMnO_4 + H_2SO_4 \rightarrow MnSO_4 + Fe_2(SO_4)_3 + K_2SO_4 + H_2O$

no. of milligram equivalent of FeO = no. of milli gram equivalent of $KMnO_4$.

$$x \times 1 = \frac{2}{5} \times 5 \times 100$$

x = 200 m moles.

Zn - dust convert all Fe^{3+} to Fe^{2+}

So no. of millimoles of Fe^{2+} in solution is = (x + 2y).

$Fe^{2+} + K_2Cr_2O_7 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + Fe^{3+} + K_2SO_4 + H_2O$

no. of milli gram equivalent of Fe^{2+} = no. of milligram equivalent of $K_2Cr_2O_7$

$$(x + 2y) \times 1 = \frac{2}{15} \times 6 \times 1000$$

$$(200 + 2y) = \frac{2}{15} \times 1000$$

y = 300 millimoles.

$$\% \text{ FeO (by moles)} = \frac{x}{x+y} \times 100 = \frac{200}{500} \times 100 = 40\%$$

% Fe_2O_3 (by moles) = 60%

Ex.35 A substance of crude copper is boiled in H_2SO_4 till all the copper has reacted. The impurities are inert to the acid. The SO_2 liberated in the action is passed into 100 mL of 0.4 M acidified $KMnO_4(SO_2 \rightarrow SO_4^{-2})$ The solution of $KMnO_4$ after passage of SO_2 is allowed to react with oxalic acid and requires 25 mL of 1 M oxalic acid. If the purity of copper is 95.25%, what was the weight of the sample.

Sol. let given sample is w gram and no. of moles of Cu in given sample is x millimoles.

$Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

x millimoles x millimoles

$SO_2 + KMnO_4 + H_2SO_4 \rightarrow MnSO_4 + H_2O + SO_3 + K_2SO_4$

no. of milligram eq. of SO_2 = no. of milli gram eq. of $KMnO_4$

x × 2 = no. of gram eq. of $KMnO_4$

left $KMnO_4$ reduced by $H_2C_2O_4$

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$KMnO_4 + H_2SO_4 + H_2C_2O_4 \rightarrow MnSO_4 + CO_2 + H_2O + K_2SO_4$

no. of milligram eq. of $KMnO_4$(left)

= no. of mill gram equ. of $H_2C_2O_4$

$= 1 \times 2 \times 25$

So, total milligram eq. of $KMnO_4$ = milligram eq. of SO_2 + milligram eq. of $H_2C_2O_4$

$0.4 \times 5 \times 100 = x \times 2 + 1 \times 2 \times 25$

x = 75 millimoles

wt(Cu) = 95.25% of W

$$75 \times 10^{-3} \times 63.5 = \frac{95.25}{100} \times W$$

$\Rightarrow$ W = 5 gram.

Ex.36 H_2O_2 is reduced rapidly by Sn^{2+}, the products being Sn^{4+} and water. H_2O_2 decomposes slowly at room temperature to yield O_2 and water. Calculate the volume of O_2 produced at 273 K and 1.00 atm when 200 g of 10.0 % by mass H_2O_2 in water is treated with 88.2 mL of 1 M Sn^{2+} and then the mixture is allowed to stand until no further reaction occurs.

Sol. Let x millimoles of H_2O_2 react with 1M, 88.2 ml Sn^{2+}

$H_2O_2 + Sn^{2+} \rightarrow Sn^{4+} + H_2O$

no. of milligram equivalent of H_2O_2

= no. of milligram eq. of Sn^{2+}

$x_{millimole} \times 2 = 1 \times 2 \times 88.2$

x = 88.2 millimoles

Total given moles of H_2O_2 is

$$n_{total} = \frac{W(H_2O_2)}{34} = \frac{10\% \text{ of } 200 \text{ gram}}{34}$$

$$= \frac{20}{34} \text{moles}$$

= 0.5882 moles

= 588.2 millimoles.

so. no. of moles of H_2O_2 that decomposed to O_2

= 588.2 – 88.2

= 500 millimoles

$$H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$$

$n(O_2)$ released = $\frac{1}{2} \times 500 = 250$ millimoles

V_{O_2} (at 1atm, 273 K) = $250 \times 10^{-3} \times 22.4 = 5.6$ litre.

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Ex.37 0.84 g iron ore containing x percent of iron was taken in a solution containing all the iron in ferrous condition. The solution required x ml of a dichromatic solution for oxidizing the iron content to ferric state. Calculate the normality of dichromatic solution.

Sol. Let Normality of $K_2Cr_2O_7$ solution is N

$Fe^{2+} + K_2Cr_2O_7 + H^+ \rightarrow Fe^{3+} + Cr^{3+} + H_2O$

no. of gram eq. of Fe^{2+} = no. of gram eq. of $K_2Cr_2O_7$

$$n_{Fe^{2+}} \times 1 = N \times x \times 10^{-3}$$

$$\frac{W_{Fe^{2+}}}{56} \times 1 = N \times x \times 10^{-3}$$

$$\frac{\frac{x}{100} \times 0.84}{56} \times 1 = N \times x \times 10^{-3}$$

$N = 0.15$

Ex.38 5 g of pyrolusite (impure MnO_2) were heated with conc. HCl and Cl_2 evolved was pssed through excess of KI solution. The iodine liberated required 40 mL of N/10 hypo solution. Find the % of MnO_2 in the pyrolusite.

Sol. Let moles of MnO_2 in 5 gram sample is x.

$MnO_2 + HCl \rightarrow MnCl_2 + Cl_2 + H_2O$

no. of gram eq. of MnO_2 = no. of gram equivalent of Cl_2

$x \times 2 = n_{(Cl_2)} \times 2$

$x = n_{(Cl_2)}$

so $n_{I_2} = n_{Cl_2} = x$

evolved Cl_2 passes through KI that released I_2

$Cl_2 + 2\,KI \rightarrow I_2 + 2KCl$

x moles x moles

Evaled I_2 required 40 ml , $\frac{N}{10} Na_2S_3O_3$

$I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

no. of gram equivalent of I_2 = no. of gram equivalent of hypo

$$x \times 2 = \frac{1}{10} \times 40 \times 10^{-3}$$

$x = 2 \times 10^{-3}$ moles

$$\%\ MnO_2 = \frac{wt.(MnO_2)}{total\ wt.} \times 100$$

$$= \frac{x \times 87}{5} \times 100 = \frac{2 \times 10^{-3} \times 87}{5} \times 100$$

$= 3.48\ \%$

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Ex.39 A 458 g sample containing Mn_3O_4 was dissolved and all manganese was converted to Mn^{2+}. In the presence of fluoride ion, Mn^{2+} is titrated with 3 lit of $KMnO_4$ solution (which was 1.25 N against oxalate in acidic medium), both reactants being converted to a complex of Mn(III) , What was the % of Mn_3O_4 in the sample ?

Sol. $KMnO_4 + H_2C_2O_4 + H_2SO_4 \rightarrow CO_2 + MnSO_4 + K_2SO_4 + H_2O$

3 litre, M molar

Normality ($KMnO_4$) = Molarity $\times$ n_f

$1.25 = M \times 5$

$M = 0.25$

Let x moles of Mn_3O_4 present in 458 gram sample

$Mn_3O_4 \rightarrow 3Mn^{2+}$

xmole 3xmoles

$Mn^{2+} + KMnO_4 \rightarrow Mn^{3+}$

no. of gram eq. of Mn^{2+} = no. of gram eq. $KMnO_4$

$3x \times 1 = (M \times n_f \times V)_{KMnO_4}$

$3x \times 1 = 0.25 \times 4 \times 3$

$x = 1$ moles

$$\% \ Mn_3O_4 \text{ (by wt)} = \frac{w(Mn_3O_4)}{w_{total}} \times 100 = \frac{1 \times 229}{458} \times 100 = 50\%$$

Ex.40 80 gm of a sample of Anhydrous $CuSO_4$ was dissolved in water and made to 250 mL. 25 mL of this solution after taking usual precaution was treated with a little excess of KI solution. A white ppt. of Cu_2I_2 and iodine was evolved. The iodine so evolved required 40 mL of 1 M hypo solution. What is the approximate purity of $CuSO_4$ solution.

[Mol. wt. of $CuSO_4$ = 160]

Sol. Let 80 gram sample contains x moles of $CuSO_4$ that desolve in 250 ml solution

So 25 ml solution contain (x/10) moles $CuSO_4$

$CuSO_4 + KI \rightarrow Cu_2I_2 + I_2$

no. of gram eq. of $CuSO_4$ = no. of gram eq. of I_2

$$\frac{x}{10} \times 1 = n_{(I_2)} \times 2$$

$$n_{(I_2)} = \frac{x}{20}$$

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Evolved I_2 required 40 ml, 1M hypo solution

$I_2 + Na_2S_2O_3 \rightarrow NaI + Na_2S_4O_6$

no. of gram eq. of I_2 = no. of gram eq. of hypo

$n(I_2) \times 2 = 1 \times 1 \times 40 \times 10^{-3}$

$$\frac{x}{20} \times 2 = 40 \times 10^{-3}$$

$x = 40 \times 10^{-2}$ moles

$$\% \ CuSO_4 \text{ (by mass)} = \frac{wt(CuSO_4)}{wt\,total} \times 100$$

$$= \frac{40 \times 10^{-2} \times 160}{80} \times 100 = 80\%$$

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